Automation, Capital Taxation, and Public Insurance

Nathan Barnard

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Abstract

Workers cannot insure automation states through assets they do not own. A government may be able to trade on their behalf, but buying more of an existing asset scales its payoff; it does not create a missing state-contingent claim. We study this distinction in a small-open economy in which automation changes wages, fiscal resources, and traded returns by different amounts. The government values a payoff by its contribution to worker continuation welfare, while international investors price the same payoff using their own stochastic discount factor. At an interior portfolio choice, the two valuations agree along the payoff that can be traded. With two shocks and one claim, an orthogonal fiscal exposure generally remains. A separate event application makes the same rank restriction visible across two possible successor economies: one inherited installed-equity position supplies one payoff vector, not two independently chosen transfers. Capital taxation acts on a different margin. It redistributes rental income and changes future capital, wages, and fiscal resources, but it cannot reproduce an untraded payoff on impact. The results characterize public insurance under restricted payoff menus; they do not provide a national-welfare ranking, a globally verified Ramsey policy, or a calibrated portfolio recommendation.

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JEL codes: E62, G11, H21, H63, O33.

1 Introduction

Automation may raise aggregate income while shifting income from labour to capital. When workers own little capital, that shift creates a natural role for policy: the government can hold claims on capital on their behalf, tax capital income, and use the proceeds to support worker consumption. Yet these policies are not interchangeable. Public ownership helps only when its returns are high in the states in which workers lose income, and the position must be financed before the shock is known. Capital taxation transfers rental income after the shock, but also changes investment, future wages, and the tax base. The question is therefore not simply whether automation raises the value of capital, but how a government should combine public asset holdings, borrowing and saving, transfers, and capital taxation to share automation's gains with workers.

This paper studies that question in economies in which workers cannot trade the relevant financial risks and the public sector has a restricted balance sheet. The central conclusion is deliberately narrow. A public claim insures only the component of worker and fiscal exposure that its payoff reaches. It can be valuable even when its expected return is modest, and it can be a poor hedge even when it pays handsomely on average. Capital-income taxation, safe borrowing, and risky claims then have different jobs: the first redistributes and changes the accumulation path, the second shifts resources across dates, and the third carries an inherited state-contingent payoff. The public problem is to combine these instruments subject to their respective payoff and incentive restrictions.

1.1 Three historical episodes and the insurance question

The episodes put three objects next to one another: worker income, returns on claims to capital, and public receipts. They do not move together. The US episode is one of unequal upside: worker resources rose alongside high returns on capital claims. The German close-up is a worker-downside episode with a capital boom. In neither case does worker incidence map mechanically into the contemporaneous position of the Treasury. These are descriptive facts, not estimates of a national automation shock; their role is to identify the distinct exposures that the model must preserve.

The United States from 1990 to 2007 provides one useful contrast. Over the Acemoglu–Restrepo robot-exposure window, the national labour share fell by roughly 0.07 percentage points per year. Aggregate real compensation nevertheless grew by about 2.35 percent per year; compensation per employee grew by about 0.9 percent and real hourly compensation by about 1.3 percent. Broad domestic equity returned about 9 percent per year in real terms. In this episode, a falling labour share is not a description of an aggregate collapse in worker resources. A public position in a capital claim would primarily share in a capital-profit boom, not mechanically insure an economy-wide wage loss.

Germany provides two different comparisons. During the 1994–2014 German robot-adjustment window, the national labour share fell by about 0.16 percentage points per year. Aggregate real compensation grew slowly, roughly 0.66 percent per year, while compensation per employee was approximately flat or slightly negative. Domestic real equity returns were nevertheless around 8–11 percent per year, depending on the long-history return construction. The 2004–07 close-up within that broader German period is sharper still. The labour share fell by about 0.97 percentage points per year; aggregate real worker compensation fell by about 0.54 percent per year; compensation per employee and real hourly compensation fell by about 1.20 and 1.51 percent per year, respectively. At the same time, the capital-tax-base proxy grew by about 3.72 percent per year and domestic real equity returned approximately 12.5–18 percent per year.

The fiscal accounts in the German close-up are equally instructive. Relative to 2000–03, total taxes in the harmonised OECD measure fell only 0.34 percentage points of GDP during 2004–07. Personal income tax and employee and employer contributions fell, while corporate income tax rose by about 0.75 points of GDP. Transfers and interest costs also fell, and the deficit improved. This is not evidence that a technology shock caused any of these movements. It is evidence against an accounting shortcut: a worker squeeze is not the same object as a contemporaneous Treasury crisis. The tax base, transfers, financing costs, and inherited public liabilities decide whether the two become aligned.

1.2 Unequal upside in the United States

The US episode helps distinguish a change in distribution from a change in levels. The labour share is the fraction of current output accruing to labour compensation. It can fall because profits and capital income grow faster than labour income even when real wages and total compensation continue to rise. That is the relevant description of the 1990–2007 window in the descriptive series: the aggregate pie and the worker component both expanded, while capital's component expanded faster. A worker who owns no capital claim may still have a reason to value a public claim on the capital boom, but the reason is participation in unequal upside, not replacement of a lost wage bill.

The distinction becomes sharper once aggregate compensation is separated from a per-worker outcome. Aggregate compensation combines compensation per employee with the number of employees and hours worked. It is informative about the resources available to workers as a group, but it is not a measure of the consumption of a representative incumbent worker. Compensation per employee and hourly compensation move more slowly in the US series. Nor do they describe the distribution across locations, occupations, or households. The displacement evidence in the automation literature is entirely compatible with a national series in which total real compensation rises. The paper's worker is therefore a stylised claimant on the wage bill and public transfers, not a claim that every worker experienced the national mean.

The equity number also needs an economic interpretation. A roughly 9 percent real annual return on broad domestic equity is a realised return on a diversified ownership claim. It combines cash distributions, price changes, and the resolution of risks that investors had priced ex ante. It is not a flow of tax revenue and it is not the payoff to a government that starts buying after the outcome is known. The relevant counterfactual is a pre-existing public exposure financed and constrained in a specified way. That timing is why the model treats a portfolio position carried into an innovation differently from a tax adjustment made after the innovation.

Figure 1 puts the episode in a longer asset-return context. It shows that domestic and external equity claims are neither constant-return claims nor interchangeable claims, and that an asset history contains long intervals unrelated to a particular automation episode. The figure therefore argues against reading a strong episode return as an unconditional case for public equity. What matters in the model is the covariance of the payoff with the worker–fiscal state, not the average historical return alone.

1.3 Worker downside and a capital boom in Germany

The broader German window is different because the aggregate and per-worker measures are much closer to one another. Slow aggregate real compensation growth together with approximately flat or slightly declining compensation per employee describes an adjustment in which the worker side of the economy did not share proportionately in the capital-income expansion. That pattern is consistent with the familiar task logic: displacement in some activities can coexist with new tasks and reallocation in others, while the incidence on the average worker remains weak. It does not reveal which mechanism dominated, and it does not identify robots as the cause of the national path. It tells us why a model should not collapse a technology transition into a single aggregate-productivity shock.

The 2004–07 close-up sharpens the accounting. In this interval, the decline in the labour share was accompanied by negative growth in aggregate real worker compensation, negative growth per employee, and negative hourly-compensation growth. Hence this is not just a compositional case in which the share fell while all worker series rose. At the same time, high domestic equity returns and growth in the capital-tax-base proxy show that the capital side can do well in a state that is adverse for the worker side. That is the textbook circumstance in which a pre-existing claim on capital might deliver resources at a valuable date. It remains insufficient to infer that a larger public equity position would have been optimal: the claim's price, the public balance sheet, and the transfer and tax system are all part of that inference.

The fiscal composition is the second lesson of the close-up. Personal taxes and social contributions move with labour income more closely than corporate-income taxes do. In the German figures, weakness in labour-linked receipts was partly offset by corporate receipts; lower transfers and interest costs also supported the fiscal position. A Treasury facing this combination may have more contemporaneous financing capacity than workers have private resources. But the result depends on the tax code, the transfer system, interest rates, debt maturity, and the initial public balance sheet. It cannot be inferred from a wage series or a corporate-profit series alone.

Figure 2 supplies a final caution about investable automation-oriented claims. The available live producer-portfolio history is short and contains both a large upside episode and a bad-leverage episode. It is useful for seeing that an investable claim can have its own financing and valuation risk. It is not long enough to establish the payoff that a public portfolio would obtain in the historical US or German episodes. The theoretical analysis accordingly begins from a specified payoff loading, not from the proposition that a labelled robotics fund is the relevant insurance asset.

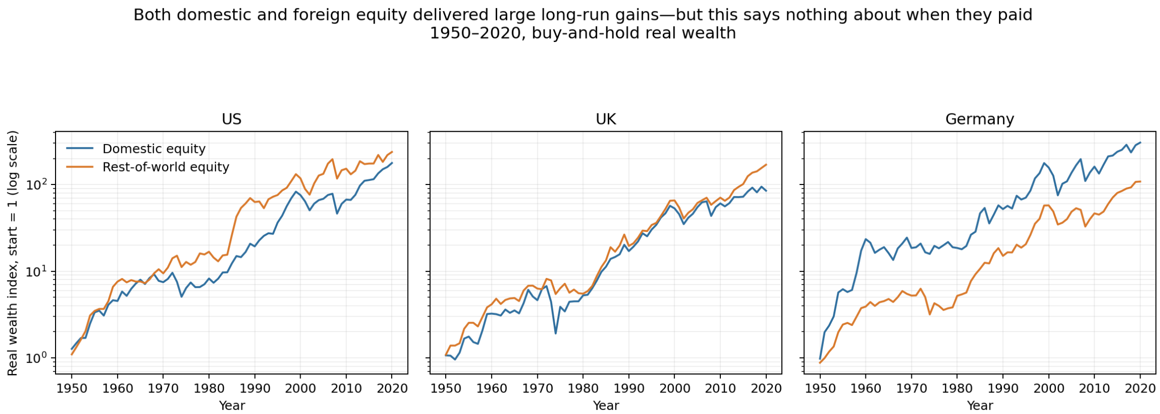

Figure 1: Long-run cumulative wealth in the descriptive asset constructions. The series are real-output illustrations of the distinct histories of broad domestic and external equity claims; they do not estimate an automation beta or a public-portfolio counterfactual. Their role is to make clear that a realised mean return is not the same object as a state-contingent public hedge.

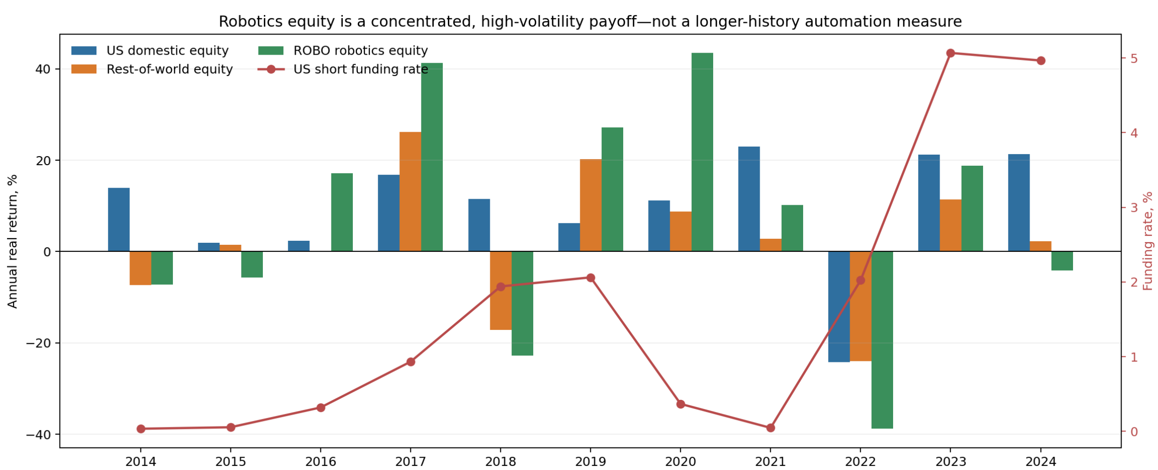

Figure 2: A short modern history of automation-oriented equity and funding conditions. This real-output descriptive figure illustrates that a tradable producer portfolio has distinct upside and financing states. It is not evidence that the portfolio replicates domestic labour income or the public tax base, and it is not used as an estimated input to the model.

1.4 From episode accounting to a fiscal-insurance model

The model below translates the historical distinctions into a small set of economic objects. The wage bill is the worker-side flow. The rental flow is the domestic capital-income flow that may be taxed. Public financial net worth is the stock that can be invested before a shock and that must be serviced after one. A risky claim has an explicitly specified payoff loading. These objects can move together in a special case, but the model does not assume that they do. The point of the formalisation is not to reproduce the episode charts; it is to show how a government should reason when the charts move in different directions.

Consider again the contrast between the US window and the German close-up. In the first, a rise in the payoff to capital can represent a source of unequal upside. A government may want a claim that shares that payoff if workers receive a smaller share of the gain than the investors who price it. In the second, the same kind of payoff can also provide resources in a worker-downside state. These are parallel insurance motives, not mutually exclusive narratives. The relevant comparison is relative: does the payoff arrive when an additional unit of public resources is valuable to workers compared with its value to the investors who supply the claim?

This comparison also explains why the worker state and the Treasury state must be distinguished. A government may be concerned with workers even if corporate receipts are high, because corporate receipts need not be transferable one-for-one to the workers at the relevant date and because tax changes affect future accumulation. Conversely, a government may have weak revenues even when workers are doing well if the tax base or financing costs deteriorate. The government kernel in the model is therefore not a reduced-form time series for total taxes. It is the shadow value of relaxing the public resource constraint in the allocation that supports worker consumption.

The historical cases also make timing unavoidable. A portfolio held before a capital boom or a worker squeeze has a payoff at the time the state is realised. A tax reform announced after the state is observed moves a flow and changes incentives over time. It may be excellent policy, but it cannot be treated as the same instrument as a pre-existing contingent claim. Similarly, a high equity valuation need not provide cash to a government that cannot sell, pledge, or finance the position at the relevant date. These timing and feasibility distinctions motivate the separate treatment of public wealth, portfolio positions, and the tax instrument in the model.

Finally, the cases suggest why it is dangerous to start from a conclusion about capital taxation. A capital-income tax can share rents associated with a technological transition. It can also discourage the accumulation that eventually restores wages and expands the tax base. The appropriate policy trades those effects off against the inherited distribution of ownership and the available claim menu. The paper develops that trade-off in a commitment Ramsey environment rather than importing a generic zero-tax conclusion or interpreting the revenue-maximising rate as the welfare-maximising one.

These episodes motivate a simple distinction that guides the paper. Let worker resources denote wages plus public transfers; let fiscal resources denote the government's financial wealth and net revenue capacity; and let a traded payoff denote the cash-flow exposure of a claim that the government can actually buy. The three are related in general equilibrium, but they are not the same object. A government can care about worker resources without being able to pledge wages. It can tax capital income without owning the capital asset. And it can hold an external risky claim whose payoff covaries with neither domestic wages nor the domestic tax base in the way that an equity label might suggest.

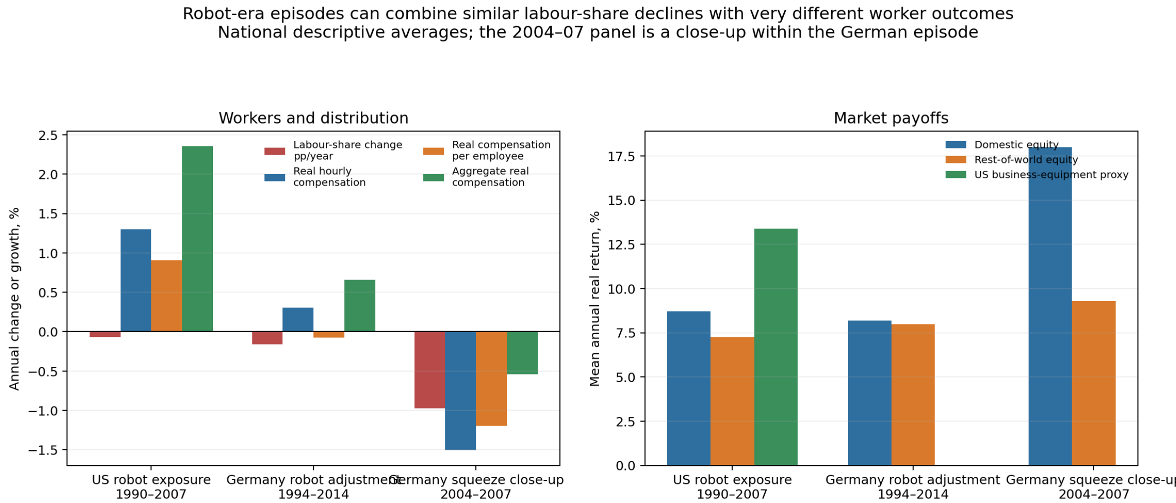

Figure 3: Worker resources and capital claims in selected US and German episodes. The series are descriptive episode summaries, not causal estimates of automation. The United States, 1990–2007, combines a declining labour share with rising aggregate real compensation and strong equity returns. Germany, 1994–2014, has weaker per-worker compensation growth; the 2004–07 German close-up combines a worker squeeze with high domestic equity returns. The Empirical Appendix describes the source series and construction.

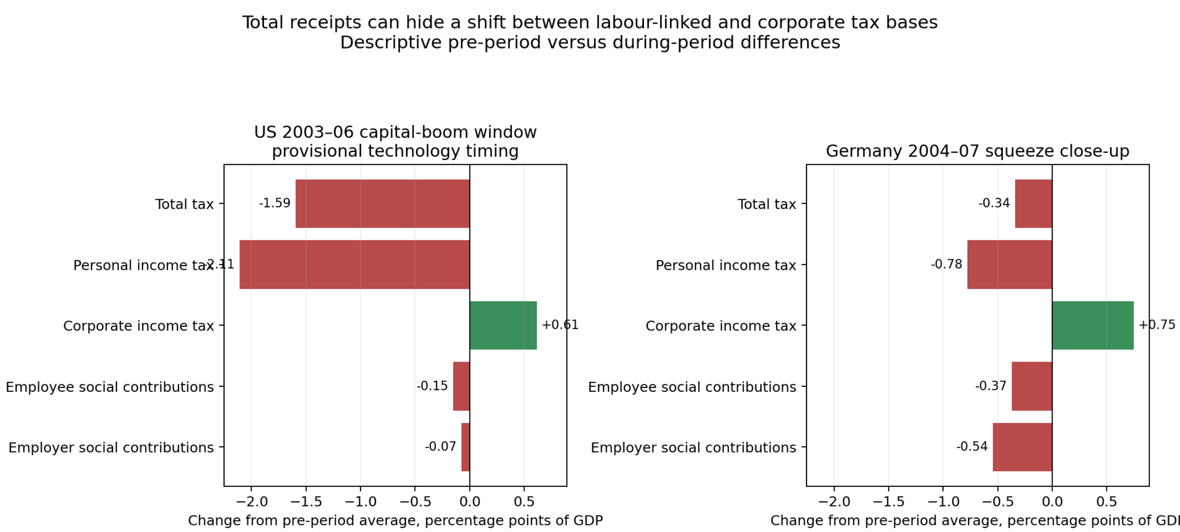

Figure 4: Tax composition in selected descriptive episodes. The figure separates total receipts from labour-linked and corporate-tax components. It is included to show why total revenue is an inadequate proxy for worker insurance or fiscal capacity; it is not an estimate of a policy or technology effect.

1.5 The economic problem

The paper asks: when workers face automation-related risk and cannot buy the relevant claims themselves, what can a commitment Ramsey government accomplish with a safe position, a restricted risky payoff menu, transfers, and a source-based tax on capital income? The question has three parts.

The first issue is valuation. Financial markets price a unit of state-contingent resources with the stochastic discount factor of the marginal international investor. The government values the same unit by the resulting relaxation of its worker-welfare problem. These values can differ because workers and the investors who set prices have different consumption exposures. A claim that pays in a state of high worker marginal utility can be valuable to the government even when private investors require little compensation for bearing that state.

The second issue is attainability. With two continuous innovations and one risky claim, the government can vary the scale of only one vector of shock loadings. It cannot select two independent contingent transfers. The unspanned component is not a technical remainder: it is the risk against which buying more of the existing claim does nothing. This is why the amount of public equity is not a sufficient statistic for public insurance.

The third issue is financing and implementation. A safe position may finance the purchase of a risky claim, but it does not create state insurance. A tax on capital income can raise revenue after a transition, yet it changes the after-tax return that governs private accumulation. The desired policy can therefore be infeasible or constrained even when its local valuation motive is clear. Fiscal wealth is a balance-sheet object; it is not a synonym for current revenue, worker income, collateral, or an unlimited debt capacity.

There is a useful asymmetry in this formulation. Public insurance can be valuable when a claim has zero market excess return, because its payoff may fill a missing worker-relevant direction at a low price. But a high expected-return claim is not automatically a fiscal resource. Its expected return compensates the marginal investor for the covariance of its payoff with that investor's marginal utility; it does not measure the transfer the government can make after taxes, portfolio bounds, and inherited debt are taken into account. The distinction is especially important when public purchases are financed by safe liabilities. Borrowing can alter the scale of a risky holding, but it cannot make the holding pay in a state where its underlying cash flow is absent.

1.6 A preview of the answer

The Brownian model makes the first two distinctions transparent. It has continuous innovations to ordinary productivity and to the range of tasks that capital can perform. Workers receive wages and transfers but do not own financial claims. Owners hold domestic installed capital and choose its accumulation. International investors price a safe asset and one external risky claim. The government chooses transfers, its safe and risky positions, and the speed at which it adjusts a capital-income tax. The external claim is not domestic installed capital: it is a separately traded, world-priced payoff. That separation prevents a public portfolio result from being silently read as a recommendation to nationalise the domestic capital stock.

The formal comparison is between the government and investor values of the same payoff. On the stated smooth interior branch, the government adjusts its risky position until the difference in those values is orthogonal to every traded return loading. Equivalently, it eliminates the component of the worker–fiscal valuation wedge that lies in the available payoff span. Equality is not state by state, and the result does not say that markets are inefficient. Different marginal claimants value resources differently; the policy problem is limited by the claims the government can trade.

This formulation encompasses two cases that are often treated separately. In a downside state, the claim pays when workers lose resources. In an unequal-upside state, workers gain but gain less than the investors who price the claim. In both cases the public value of the payoff can exceed its market value. Conversely, an automation innovation may move the wage bill and tax base while leaving the claim unaffected. This claim-neutral negative control is central: no portfolio scale can cover a direction in which the claim has zero loading.

The portfolio result does not settle capital taxation. A pre-existing risky position has an impact payoff when an innovation arrives. The tax rate and installed capital are inherited at that instant. Changing the tax rate redistributes rental income and changes the future accumulation path, but it cannot manufacture a missing impact payoff. The long-run tax calculations therefore ask a different question: how does a permanent wedge trade off current redistribution against the future wage, output, and tax-base consequences of slower capital accumulation? In the log-utility benchmark, the one-half limiting tax arises from the common-return balance between private accumulation and fiscal valuation, not from a generic rule that capital taxation should be zero or that revenue should be maximised.

The event-state model changes the geometry rather than merely adding another shock. A technological arrival can lead to a partial-automation successor economy or a limiting AK successor economy in which the conventional labour-income channel disappears. One inherited installed-equity position then produces a pair of successor-state payoffs. Changing one scalar position moves public wealth along that pair; it does not allow the government to choose successor wealth independently in the two economies. The distinction between physical arrival probabilities and risk-neutral pricing weights is therefore essential. The former govern resource feasibility and expected welfare; the latter price a traded payoff. The model is a conditional analysis of successor production structures, not a forecast of a date or probability of transformative AI.

1.7 Contributions and relation to the literature

The paper contributes a fiscal-insurance perspective to three literatures. First, it extends the public-portfolio logic in Aparisi de Lannoy et al. (2025). The relevant statistic is not a claim's expected excess return, but the state-contingent value of its payoff to a government facing worker needs and fiscal constraints. The additional issue here is that automation can change both factor incomes and the set of contingent payoffs a government can reach. A public menu that spans some fiscal exposure does not thereby complete household markets or implement the first best.

Second, the paper is related to incomplete-markets Ramsey work in which debt, assets, and taxes are joint instruments. Bhandari et al. (2017) show why state-contingent fiscal needs connect debt management to tax policy, while Farhi (2010) connects capital taxation to ownership when markets are incomplete. Our contribution is to keep the payoff-span and incidence restrictions explicit in an economy where a technology change affects worker income, capital accumulation, and the tax base at the same time. This makes it possible to say precisely what a public claim accomplishes and what it leaves uncovered.

Third, the paper builds on task-based accounts of automation. Acemoglu and Restrepo (2019) emphasise displacement and reinstatement across tasks, and Moll et al. (2022) show how automation can generate uneven income and wealth dynamics. We use task production to derive worker and fiscal exposure rather than assume a reduced-form wage loss. Work on advanced AI and long-run growth explores circumstances in which the role of labour could change more radically (Aghion et al., 2019; Korinek and Suh, 2024; Restrepo, 2025; Trammell and Korinek, 2026; Jones, 2026; Jones and Tonetti, 2026). Our AK successor is a transparent limiting case within that broader discussion. It does not select among those forecasts. The paper instead asks what the public balance sheet can insure if the successor production regime is uncertain.

1.8 Road map and scope

Section 2 lays out the Brownian economy from household choices to the government budget and competitive equilibrium. Section 3 states the commitment Ramsey problem. Sections 3 and 4 derive the valuation and payoff-span implications. Section 5 studies taxes, accumulation, and the local linear–quadratic benchmark. Section 6 turns to successor economies and the AK limit. Section 7 reports the fixed-policy transition accounting exercise, and the final section draws out the implications.

The paper distinguishes analytic results from illustrative calculations throughout. The episode figures in this section are provisional descriptive evidence. The numerical exercises are diagnostic solutions or fixed-rule accounting exercises, not empirical estimates or global policy prescriptions. These boundaries are substantive: the paper's contribution is to clarify the economic objects that would be required for a quantitative policy evaluation, rather than to attach false precision to an unsettled calibration.

2 The economy

The economy separates a worker's wage claim, an owner's installed capital, and an external claim traded at world prices. This separation makes a public portfolio and a source-based capital-income tax distinct instruments. We describe the private economy first and then define the competitive equilibrium that constrains the Ramsey plan.

2.1 Time, uncertainty, and inherited positions

Time is continuous, $t \geq 0$. The economy is small in world capital markets. There are two Brownian innovations, collected in $dB_t = (dB_t^z, dB_t^a)'$, with $d\langle B^z, B^a \rangle_t = 0$. The first moves ordinary total factor productivity. The second moves the range of tasks that can be performed by capital. We write

$$dz_t = \mu_z(z_t)dt + \sigma_z(z_t)dB_t^z, \qquad da_t = \mu_a(a_t)dt + \sigma_a(a_t)dB_t^a, \tag{2.1}$$

where $a_t \in (0, 1)$ is the share of tasks that are technically automatable. The drifts and volatilities are primitives. They may be estimated or calibrated in a quantitative application, but the results below do not interpret them as an identified historical automation process.

At date zero the economy inherits physical and financial states

$$S_0 = (z_0, a_0, K_0, N_0, \tau_0). \tag{2.2}$$

Here K_0 is installed domestic capital, N_0 is public financial net worth, and τ_0 is the source-based capital-income tax rate. The baseline starts from these physical, financial, and policy states and contains no additional protected promise.

The timing distinction is important. Installed capital and the tax rate have continuous paths. A risky position chosen before an innovation, in contrast, has an immediate Brownian payoff. Thus a public portfolio determines impact exposure; tax adjustment and capital accumulation determine part of the response after impact. The distinction is not a statement that taxes are useless. It identifies the date at which each instrument can move resources.

2.2 Tasks, firms, and factor prices

There is a unit mass of tasks. A fraction a_t can be performed with capital and the remaining fraction requires labour. Competitive firms combine installed capital K_t and a fixed aggregate labour supply $L > 0$ to produce final output. On the full-specialisation branch used in the main text, the task-allocation problem aggregates to

$$Y_t = Z_t \Omega(a_t) K_t^{a_t} L^{1-a_t}, \qquad Z_t > 0, \tag{2.3}$$

where Z_t is the level of ordinary productivity, with $\log Z_t = z_t$, and

$$\Omega(a) = a^{-a}(1-a)^{-(1-a)}. \tag{2.4}$$

The expression is the familiar allocation-adjusted Cobb–Douglas representation of the task block. It is not assumed to be globally valid for every possible state. The Theory Appendix derives it from the task allocation and records the inequalities that define the branch. Keeping those conditions out of the main text lets the reader see the economic incidence without confusing a local production region with a claim about all technologies.

Equalizing factor use across the tasks assigned to each factor produces the $\Omega(a_t)$ adjustment. The Theory Appendix gives the allocation problem and the inequalities defining this specialization branch.

Given factor prices and technology, a representative competitive firm chooses factor services $\widetilde{K}$ and $\widetilde{L}$ according to

$$\max_{\widetilde{K},\widetilde{L}} \left\{ Z_t \Omega(a_t) \widetilde{K}^{a_t} \widetilde{L}^{1-a_t} - R_t^K \widetilde{K} - w_t \widetilde{L} \right\}. \tag{2.5}$$

In aggregate, installed K_t and inelastic labour supply L are predetermined factor supplies, and market clearing sets factor demand equal to those supplies. Euler's theorem for the constant-returns technology implies that the two factor payments exhaust output. Differentiating the production function gives (2.6); multiplying the wage by L and the rental rate by K gives $W_t = (1-a_t)Y_t$ and $R_t^K K_t = a_t Y_t$. Depreciation is paid out of the capital claim rather than from the firm's gross marginal product, which explains the net rental base in (2.7). This derivation also shows why a source tax applies to the rental flow rather than to the market value of capital. The latter embeds the value of all future rentals, whereas the tax is levied on the current flow.

Profit maximisation gives the rental rate on installed capital and the wage per unit of labour,

$$R_t^K = a_t \frac{Y_t}{K_t}, \qquad w_t = (1 - a_t)\frac{Y_t}{L}. \tag{2.6}$$

The wage bill and signed net rental tax base are therefore

$$W_t = w_t L = (1 - a_t) Y_t, \qquad \mathcal{B}_t = (R_t^K - \delta) K_t, \tag{2.7}$$

where δ is depreciation. These definitions separate three quantities that are too often conflated. R_t^K is a domestic rental rate, not the world safe rate; $\mathcal{B}_t$ is taxable rental income, not a traded security payoff; and W_t is private labour income, not a public asset.

Holding K_t and L fixed, the local incidence of a change in automation is

$$d \log Y_t = d \log Z_t + s_a(a_t) da_t, \qquad d \log W_t = d \log Z_t + \left[s_a(a_t) - \frac{1}{1 - a_t} \right] da_t, \tag{2.8}$$

where $s_a(a) = \partial \log[\Omega(a) K^a L^{1-a}]/\partial a$ holds the stocks fixed. The first expression contains the scale effect of a larger automatable task set. The second also contains the fall in labour's task share. Output can consequently rise while the wage bill falls. It can also raise both. The model needs neither sign as a maintained empirical fact; it needs the possibility that worker resources, the tax base, and asset payoffs move differently.

2.3 Automation and factor incidence

Suppose that the range of capital-performed tasks expands while the stock of installed capital is fixed. The economy receives a productive scale effect because a larger set of tasks can be performed with the available capital. But labour is assigned to the remaining tasks, whose measure has fallen. The two effects appear separately in (2.8). This is the basic reason that aggregate output is not an adequate statistic for the worker-insurance problem.

The same representation keeps several otherwise hidden margins in view. An expansion of a_t can change the rental rate, the wage bill, and the return to retaining installed capital before it changes the capital stock itself. A permanent movement also changes the incentive to accumulate, so the short-run and long-run wage responses need not have the same sign. In particular, an automation advance can initially lower the wage bill through the task-share channel and later raise it after capital expands. The later tax analysis exploits this timing. It does not infer the transition path from a static factor-share comparison.

The task block also clarifies the status of the full-automation limit used later. As a_t approaches one, the conventional labour-task measure shrinks. The AK successor economy does not arise by mechanically substituting one into (2.3); it is a separately specified successor technology with reproducible capital and no ordinary wage bill. Keeping the limiting economy separate avoids treating a boundary of a local Brownian approximation as a complete description of a different production regime.

2.4 Workers

Workers supply the inelastic labour quantity L. They do not hold domestic capital, the safe asset, or the external risky claim. This segmentation is a modelling device that creates a role for public intermediation; it should not be read as a literal description of every household's portfolio. A worker receives the wage bill and a non-negative transfer T_t and consumes

$$c_t^W = W_t + T_t, \qquad T_t \geq 0. \tag{2.9}$$

There is no Euler equation for workers because they have no saving vehicle in this environment. The government evaluates allocations with

$$U^W = \mathbb{E}_0 \int_0^\infty e^{-\rho_W t} \log c_t^W \, dt, \qquad \rho_W > 0. \tag{2.10}$$

Worker welfare does not imply that wages are pledgeable public collateral. Wages matter because they determine the transfer that is needed to sustain worker consumption. The government cannot borrow against them unless a separate contract explicitly permits that operation.

If workers could trade the same claims at world prices, the public problem would concern initial distribution rather than intermediation of this missing market. Here transfers are the only way workers receive the state-contingent resources carried by the public balance sheet.

2.5 Owners and capital accumulation

Domestic owners hold the installed capital stock and receive the after-tax rental flow. With logarithmic preferences and the maintained reversible-installation normalization, an owner solves

$$\sup_{\{C_t^K\}} \mathbb{E}_0 \int_0^\infty e^{-\rho_K t} \log C_t^K \, dt \quad \text{subject to} \quad \dot{K}_t = (1 - \tau_t)(R_t^K - \delta)K_t - C_t^K. \tag{2.11}$$

With a unit price of installed capital, the current-value costate condition and the logarithmic consumption condition give

$$C_t^K = \rho_K K_t. \tag{2.12}$$

Substituting this rule into the owner's budget gives

$$\dot{K}_t = K_t \left[(1-\tau_t)(R_t^K - \delta) - \rho_K\right]. \tag{2.13}$$

The capital law is therefore the private optimal response to the rental flow and tax wedge. The government internalises its consequences for workers, but it does not replace the owner with a social planner.

The owner first-order condition also makes the role of the tax wedge precise. Let q_t be the current-value shadow value of installed capital. The consumption condition is $q_t = 1/C_t^K$, while the costate equation equates the return on retaining one additional unit of capital with the required return on wealth. Under the unit- price normalization and log preferences these conditions imply a constant consumption- wealth ratio, $C_t^K/K_t = \rho_K$. The tax enters the law of motion through $(1-\tau_t)(R_t^K - \delta)$, not through the investor's discount factor. A permanent increase in τ_t therefore has an immediate distributional effect on the rental flow and a dynamic effect on K_t. The latter is the channel through which an otherwise attractive source tax can reduce future wages and fiscal revenue.

The government does not choose K_t directly. It influences accumulation by changing the after-tax rental return. A higher capital-income tax can redistribute a larger share of the existing rental flow while the stock remains in place, but it lowers the return that supports subsequent accumulation. This is the primitive trade-off behind the tax results later in the paper.

The owner block is intentionally parsimonious. It keeps ownership of domestic installed capital separate from the external risky claim traded by the government. It also makes the transition channel legible: a tax change affects future wages and fiscal resources through K_t, rather than through an assumed direct public control of investment. Giving the government a direct investment technology would be a different Ramsey problem.

2.6 World investors and asset prices

International investors supply a safe asset with instantaneous return r_t and price an external risky claim. Let m_t^I be their stochastic discount factor:

$$\frac{dm_t^I}{m_t^I} = -r_t dt - \lambda'_{I,t} dB_t, \tag{2.14}$$

where $\lambda_{I,t} = (\lambda_{Iz,t}, \lambda_{Ia,t})'$ is the vector of world prices of the two innovations. Let R_t denote the total return on the external claim, with loading vector $\sigma_{R,t}$. Its no-arbitrage return equation is

$$dR_t = \left(r_t + \sigma'_{R,t}\lambda_{I,t}\right) dt + \sigma'_{R,t} dB_t. \tag{2.15}$$

Equivalently, the discounted gain on any traded self-financing claim is a martingale under the pricing measure induced by m^I. The claim's risk premium is a price for exposure borne by the investors who set m^I. It is not, by itself, evidence about the claim's value for workers.

The pricing equation can be read in three steps. The safe rate r_t is the payment for deferring one unit of resources without Brownian risk. The loading vector $\sigma_{R,t}$ says how much the claim's payoff moves with productivity and the task innovation. The inner product $\sigma'_{R,t}\lambda_{I,t}$ is the compensation that international investors require for accepting that combination of exposures. A positive exposure to a state in which their marginal utility is high carries a corresponding price. But the government cares about worker consumption and the shadow value of public wealth, so its valuation kernel need not equal m^I.

This point does not require a belief that international investors make an error. The same dated, contingent unit can rationally have different marginal values for agents with different endowments and different constraints. The government cannot purchase the unit at its own valuation; it must purchase it at the world price. The portfolio problem later asks whether it is worth paying that price for a payoff that relaxes the worker-welfare problem. This is the same distinction that separates actuarial likelihood from a state price: a likely state need not be cheap, and a bad worker state need not be a bad state for the investor who prices the claim.

The term "external" does real work. The claim is not a share of K_t, not a claim on the domestic tax base, and not an accounting claim on future wages. Its relation to domestic technology is encoded only through its return loading σ_R. The government may find an external claim useful because it pays in a worker-relevant state, even though neither its issuer nor its cash flows are domestic. Conversely, domestic capital can appreciate strongly while the external claim delivers little insurance. This is why the later analysis is formulated in payoffs rather than security labels.

2.7 Public instruments and the flow budget

The government chooses worker consumption c_t^W, the dollar position s_t in the external risky claim, and a tax-adjustment speed ν_t. Its residual position $N_t - s_t$ is held in the safe asset. The source-tax rate follows

$$d\tau_t = \nu_t dt, \qquad \tau_t \leq 1, \tag{2.16}$$

and changing it uses final goods according to

$$\Psi_{\tau,t} = \frac{Y_t}{2\kappa_\tau}\nu_t^2, qquad\kappa_\tau > 0. \tag{2.17}$$

The control is the speed of adjustment rather than the tax rate itself. At a tax boundary, the admissible direction of ν_t is correspondingly one-sided.

Public financial net worth obeys

$$dN_t = \left[(N_t - s_t)r_t + s_t\big(r_t + \sigma'_{R,t}\lambda_{I,t}\big) + \tau_t\mathcal{B}_t + W_t - c_t^W - \Psi_{\tau,t}\right]dt \\ + s_t\sigma'_{R,t}dB_t. \tag{2.18}$$

Using the worker budget in (2.9), the domestic fiscal flow in brackets is simply $\tau_t\mathcal{B}_t - T_t - \Psi_{\tau,t}$. Writing the budget in terms of c_t^W is convenient because the government chooses worker consumption; it does not mean that wages are revenue. Likewise, N_t is financial net worth, not the present value of future statutory tax receipts. A continuation solvency requirement imposed below is a separate restriction on feasible policy.

The diffusion term in (2.18) contains the payoff restriction that drives the paper. One scalar position s_t scales the two elements of $\sigma_{R,t}$ together. The government cannot choose its exposure to productivity and task automation independently unless it has another linearly independent risky claim. Tax changes may alter transfers and the future capital stock, but they create no instantaneous Brownian payoff in a direction absent from σ_R.

Each term in the public budget has a distinct interpretation. $(N_t - s_t)r_t$ is the return on the residual safe position. The next term is the expected return on the external risky position, including its compensation for world-priced risk. The tax term is current revenue from the domestic rental flow. The expression $W_t - c_t^W$ is minus the transfer once the worker budget is imposed; retaining the longer expression makes clear that the government chooses consumption rather than confiscating wages. Finally, $\Psi_{\tau,t}$ is a real adjustment cost, not a financial transfer between domestic agents. It absorbs goods and hence appears both in the government budget and the resource constraint.

The stochastic component is equally important. The safe position has no Brownian loading. The risky position contributes $s_t \sigma_{R,t}$, and the government can change only the scalar s_t. A large position can therefore carry a large gross return demand and a nearly offsetting hedge demand. Observing a small net position does not establish that risk is unimportant, just as observing a large net position does not establish that it is well aligned with worker exposure. The decomposition in Section 3 makes this observation formal.

2.8 Feasibility, clearing, and equilibrium

Feasibility requires $K_t > 0$, $c_t^W > 0$, and the transfer and tax restrictions just stated. The government is also subject to portfolio bounds and a continuation solvency condition. Rather than build a particular debt limit into the primitive budget identity, let $\mathcal{A}(S_t)$ denote the admissible set of safe and risky positions, tax speeds, and transfers at state S_t. This notation keeps two distinct ideas separate: a numerical borrowing constraint is a policy restriction; an intertemporal no-Ponzi condition rules out financing plans that cannot be continued. The exact admissibility conditions used in each analytical and numerical exercise are stated with that exercise.

This separation is economically useful. A portfolio bound may reflect an institutional limit on sovereign leverage or a restriction on short selling a public claim. It limits which payoff directions the government can purchase even when the present value of future surpluses is large. A tax bound limits how quickly or how far the government can redistribute rental income. The transfer floor prohibits financing a balance-sheet operation by making workers' consumption negative. And solvency requires that a proposed continuation finance its commitments under the maintained prices and policy rules. None can be replaced by a statement that the government has a large tax base.

The distinction between capacity and valuation is especially important at a boundary.

Suppose a claim pays in a state in which worker marginal utility is high. The government may value an additional unit of that claim highly, yet be unable to increase its holding because safe borrowing is bounded or because future transfers cannot support the servicing cost. Conversely, a government may have room to borrow but no claim whose payoff reaches the worker-relevant innovation. The later first-order conditions are therefore interior characterisations. When a constraint binds, the associated multiplier is part of the economic answer rather than a numerical detail.

The domestic resource constraint allows for the current-account flow associated with the public sector's foreign asset position. Writing NX_t for net exports, it is

$$Y_t + NX_t = C_t^K + c_t^W + \dot{K}_t + \delta K_t + \Psi_{\tau,t}. \tag{2.19}$$

The current account is the real counterpart of changes in foreign claims implied by the public budget. The country is small, so the rest of the world takes the other side of its safe and risky positions at the prices in (2.14)–(2.15). There is no domestic market-clearing condition requiring public holdings of the external claim to equal the domestic capital stock.

The equilibrium conditions can now be read as a sequence. The two state processes determine the current technology. Given the state and installed capital, firm optimisation determines output, wages, and the rental flow. The worker budget turns wages and transfers into worker consumption. Owner optimisation turns the after-tax rental flow into consumption and next period's installed capital. World pricing determines the price and expected return on the external claim. Finally, the public budget determines the evolution of financial net worth implied by the government's chosen transfers, portfolio, and tax adjustment. Goods and foreign-asset clearing ensure that these individual decisions add up. This order is useful because it makes clear where policy has direct control and where it operates only through equilibrium responses.

Definition 2.1 (Competitive equilibrium for a given policy)**.** Fix an initial state S_0 and an admissible government policy $\{c_t^W, s_t, \nu_t\}_{t\geq 0}$. A competitive equilibrium is a collection of state processes $\{z_t, a_t, K_t, N_t, \tau_t, Y_t, w_t, R_t^K, W_t, \mathcal{B}_t\}_{t\geq 0}$ and prices $\{r_t, \lambda_{I,t}, R_t\}_{t\geq 0}$ such that: (i) the exogenous technology states follow (2.1); (ii) firms maximise profits and factor prices satisfy (2.6); (iii) workers satisfy (2.9); (iv) owners satisfy (2.12)–(2.13); (v) world prices satisfy (2.14)–(2.15); (vi) the public budget (2.18), goods clearing (2.19), and all admissibility conditions hold.

The definition makes clear what policy takes as given. The government does not choose the world price of risk, the external claim's loading, or private capital directly. It chooses policies subject to the competitive allocation they induce. This is the appropriate constraint set for a commitment Ramsey problem.

2.9 From competitive equilibrium to Ramsey policy

At date zero, the government chooses an admissible contingent policy and its induced competitive equilibrium to maximise (2.10), taking inherited physical and financial states—and any protected commitments where they are present—as given. The next section writes that commitment Ramsey problem formally. It is important to state it only after the household, firm, pricing, and government conditions are on the table: the government's value of an extra public dollar depends on the private allocation and the financial prices that the policy cannot simply declare away.

2.10 Three benchmarks

The economy nests three useful benchmarks. First, suppose the task state is locally constant, $\sigma_a = 0$. The model then has ordinary productivity risk but no automation-risk innovation. A public portfolio can still have a fiscal role, but the special question of insuring a change in the task allocation disappears. This benchmark separates a general public-portfolio motive from the mechanism associated with automation.

Second, suppose the government has no risky claim, so $s_t = 0$ is imposed. Safe assets, transfers, and the capital-income tax remain. The government can smooth deterministic resources across dates and redistribute the rental flow, but it has no instrument that delivers an impact payoff contingent on either Brownian innovation. Comparing this case with the baseline isolates what a risky public claim adds; it does not say that safe debt is irrelevant for fiscal feasibility.

Third, suppose workers and the marginal investor have proportionate exposure to both innovations, after accounting for the government's resource constraint. The relative value of an incremental payoff then has no Brownian component. The fiscal hedge motive vanishes even if the claim earns a risk premium and even if the government continues to use it for other reasons. This benchmark is useful because it shows that the paper's result is not driven by an assertion that all risky claims should be held publicly. The motive is the difference between worker-fiscal and investor exposure, projected onto a claim the government can trade.

3 Commitment policy and fiscal valuation

The government chooses policy at date zero and remains committed to the resulting contingent plan. Its objective is worker welfare. Workers do not trade; domestic owners save and own productive capital; international investors price the external claim. The government can redistribute current rental income, shift resources across dates, and buy exposure to the two aggregate innovations. These instruments enter one budget, but they do not perform the same economic task.

The Ramsey problem is easiest to understand in two stages. First, take the competitive equilibrium described in Section 2 as an implementation constraint. Second, ask how the planner values a marginal dollar delivered in a particular date and state. The value need not equal the market price of that dollar, because the marginal international investor and the workers represented by the government have different consumption exposures. Public trading exploits this valuation difference only along the payoffs the government can actually buy.

3.1 The commitment Ramsey problem

Let $S_t = (z_t, a_t, K_t, N_t, \tau_t)$ collect productivity, automation, installed capital, public financial net worth, and the capital-income tax rate. The inherited state S_0 is fixed when the plan is chosen. A policy consists of worker consumption c_t^W, the market value s_t of the external risky position, and the tax-adjustment rate $\nu_t = \dot{\tau}_t$. The government chooses an admissible contingent policy and the competitive equilibrium it induces to solve

$$V(S_0) = \sup_{\{c_t^W, s_t, \nu_t\}_{t \geq 0}} \mathbb{E}_0 \int_0^\infty e^{-\rho_W t} \log c_t^W \, dt, \tag{3.1}$$

subject to the household problems, firm first-order conditions, asset-pricing equations, public budget, market clearing, policy bounds, and transversality conditions stated in Section 2. Calling this a commitment Ramsey problem fixes the timing: after an innovation, the economy follows the contingent continuation chosen at date zero. It does not give the government an opportunity to confiscate installed capital or rewrite protected private claims at the instant the state changes.

On a smooth interior continuation the recursive representation is

$$
\begin{aligned}
\rho_W V = \max_{c^W, s, \nu} \Bigg\{ & \log c^W + \mathcal{L}^{z,a} V + K g V_K + \nu V_\tau \\
& + \left[rN + s\beta_R + \tau \mathcal{B} + W - c^W - \frac{Y}{2\kappa_\tau} \nu^2 \right] V_N \\
& + s D[V] + \frac{1}{2} s^2 \|\sigma_R\|^2 V_{NN} \Bigg\},
\end{aligned}
\tag{3.2}
$$

where $\mathcal{L}^{z,a}$ is the generator of the two technology states, $g = \dot{K}/K$ is private capital growth, $\mathcal{B}$ is the taxable rental base, $\beta_R \equiv \sigma_R' \lambda_I$ is the claim's expected excess return under the market price of risk, and

$$
D[V] = \sigma_z \sigma_{Rz} V_{Nz} + \sigma_a \sigma_{Ra} V_{Na}. \tag{3.3}
$$

The first line of (3.2) propagates technology, capital, and the tax state. The second values public cash flow. The third records the covariance between the external payoff and the marginal fiscal value of wealth. This grouping is the economic content of the recursion; the coordinate transformations used to solve it are confined to the appendices.

The consumption and tax-speed conditions on the same interior branch are

$$
c^W = \frac{1}{V_N}, \qquad \nu = \frac{\kappa_\tau}{Y} \frac{V_\tau}{V_N}. \tag{3.4}
$$

The first says that the shadow value of one more public dollar is worker marginal utility. The second says that tax adjustment balances the shadow value of moving the tax state against its resource cost. At a transfer floor, tax bound, position limit, or solvency boundary these equalities give way to the corresponding one-sided condition.

3.2 The portfolio trade-off before asset-pricing notation

Here V is the worker-welfare continuation value and V_N is the marginal worker value of one additional unit of public net worth. Let c_t^W be worker consumption, s_t the dollar market value of the public external risky holding, and $\nu_t = \dot{\tau}_t$ the tax-adjustment speed. A policy is evaluated by expected discounted worker log consumption, subject to the sequential private and public budgets, market clearing, and the stated continuation restrictions. On the smooth interior branch considered here, an extra dollar of public net worth has shadow value $V_N > 0$, while additional public wealth is locally diminishing in value, $V_{NN} < 0$. The portfolio faces three forces:

1. The claim earns the market excess return β_R.
2. Its payoff moves public net worth in particular productivity and automation states, changing the marginal value of transfers and future fiscal resources.
3. Its own risk is costly when the continuation value is concave in public net worth.

These forces enter the same marginal decision. To see the relevant direction without introducing a matrix, write the return loading as

$$\sigma_R = (\sigma_{Rz}, \sigma_{Ra})', \qquad dR_t = (r_t + \beta_R)dt + \sigma_{Rz} dB_t^z + \sigma_{Ra} dB_t^a. \tag{3.5}$$

The first component is exposure to the productivity innovation and the second to the automation innovation. A change in s can alter public exposure only in the ratio $(\sigma_{Rz}, \sigma_{Ra})$. It cannot independently choose two state transfers.

The HJB characterizes a smooth interior candidate. The theory appendices state the admissibility, boundary, and transversality conditions needed to distinguish this local characterization from a globally feasible and unique Ramsey allocation.

3.3 A two-state version of the insurance problem

Before returning to continuous time, consider a single future date with two states, d and u. A claim costs q today and pays (R_d, R_u) tomorrow. Let (q_d^I, q_u^I) denote the market state prices implicit in q, and let (q_d^G, q_u^G) denote the government's marginal fiscal values of a dollar in the same states. Buying s units changes the objective at the margin by

$$R_d(q_d^G - q_d^I) + R_u(q_u^G - q_u^I). \tag{3.6}$$

The government buys the claim when its payoff is tilted toward states in which a public dollar is worth more than its market cost. At an interior optimum, the expression is zero after taking account of how the position itself changes marginal fiscal values.

This elementary condition already separates expected return from insurance. A claim can have a high average payoff because it performs badly in states that the marginal market investor values highly. That premium is compensation, not free resources. The same claim insures workers only if its payoff is high when the fiscal value of a dollar is high relative to the market value. Conversely, a zero-premium claim can be valuable to the government if it transfers resources across states that workers and market investors value differently.

The two states need not mean absolute worker loss and gain. In the downside case, automation lowers worker consumption in state d, so the government values resources there especially highly. In the unequal-upside case, worker consumption rises in both states, but rises less than international-investor consumption in d. Marginal worker utility then rises relative to market marginal utility in d, and the same fiscal insurance motive appears. This is why the paper is not built on the premise that automation must make workers absolutely poorer. It is built on disagreement over the state in which one more public dollar is valuable.

With a single claim, equation (3.6) supplies only one linear restriction. The position cannot generally make $q_d^G = q_d^I$ and $q_u^G = q_u^I$ separately. A second linearly independent payoff would be needed for that stronger state-by-state alignment. The Brownian analysis below is the instantaneous analogue: the two states become two innovation directions, and the claim payoff becomes a vector of diffusion loadings.

3.4 Market and fiscal values of the same payoff

The market SDF prices the traded claim. Under the small-open pricing closure,

$$\frac{dm_t^I}{m_t^I} = -r_t dt - \lambda_{Iz,t} dB_t^z - \lambda_{Ia,t} dB_t^a, \qquad \beta_{R,t} = \sigma_{Rz,t}\lambda_{Iz,t} + \sigma_{Ra,t}\lambda_{Ia,t}. \tag{3.7}$$

Thus a payoff is expensive when it arrives in a state in which marginal international consumption is valuable. Equation (3.7) does not say that the payoff insures workers or relaxes a fiscal constraint.

Along a candidate Ramsey continuation, the fiscal marginal value of a public dollar is represented by

$$m_t^G = e^{-\rho_W t} V_N(S_t). \tag{3.8}$$

This is a fiscal valuation kernel induced by the worker-welfare problem. It is not an additional household SDF and does not price unavailable claims. If its diffusion is written with the same sign convention as the market kernel, then

$$\left.\frac{dm_t^G}{m_t^G}\right|_{dB} = -\lambda_{Gz}(s) dB_t^z - \lambda_{Ga}(s) dB_t^a. \tag{3.9}$$

The ratio

$$\Theta_t = \frac{m_t^G}{m_t^I} \tag{3.10}$$

therefore rises after a Brownian innovation when a public dollar becomes more valuable relative to its market price. Its level concerns marginal valuations at the current state; its drift concerns intertemporal valuation and safe finance; its diffusion is the state-contingent object relevant for the risky holding.

Itô's formula makes the portfolio channel concrete. Let the diffusion of V_N due to primitive states be $(\sigma_z V_{Nz}, \sigma_a V_{Na})'$. The claim adds the two components $sV_{NN}\sigma_{Rz}$ and $sV_{NN}\sigma_{Ra}$. Hence

$$\begin{aligned}\lambda_{Gz}(s) &= -\frac{\sigma_z V_{Nz} + sV_{NN}\sigma_{Rz}}{V_N},\\ \lambda_{Ga}(s) &= -\frac{\sigma_a V_{Na} + sV_{NN}\sigma_{Ra}}{V_N}.\end{aligned} \tag{3.11}$$

The public position moves both fiscal risk prices in the single proportion dictated by the claim payoff. This is the decisive restriction behind the paper's payoff-span argument.

3.5 From worker marginal utility to the fiscal discount factor

The fiscal discount factor can be derived directly from the transfer margin. On the interior branch, an extra unit of worker consumption costs one unit of public resources. The consumption first-order condition in (3.4) therefore sets $V_N = 1/c$. Moving along the Ramsey continuation and discounting at the worker rate gives m_t^G. Its innovation reflects every way in which the state changes the value of a public dollar: the direct transfer it can finance, the future wage path, the future capital-income tax base, and the possibility that a fiscal constraint becomes tight.

In the simplest log-consumption benchmark this statement reduces to a familiar ratio. If m_t^I is also represented by international-investor marginal utility, then the relative fiscal value is proportional to the ratio of international to worker consumption. A state in which worker consumption falls relative to investor consumption makes the fiscal discount factor rise relative to the market discount factor. The full Ramsey model adds the endogenous continuation through capital, taxes, and public net worth, but it does not change the basic comparison.

Three distinctions are important. First, the *level* of the relative discount factor measures the current marginal valuation of public resources. Its innovation is the object relevant for a risky payoff. A high level of inequality without state variation need not create a portfolio motive. Second, m^G is a shadow valuation kernel. It prices the marginal payoff for the government; it is not a claim that workers trade the payoff themselves. Third, optimal trading equates fiscal and market valuation only along attainable payoffs. It does not make m^G a market pricing kernel for every untraded state.

3.6 Safe timing and risky-state insurance

The safe position and the risky claim act on different parts of the relative valuation problem. Safe borrowing or saving moves one unit of resources from one date to another. Its first-order condition concerns the drift of the relative discount factor: whether the fiscal system values an additional future dollar more or less than its market price. The risky claim changes resources across contemporaneous innovation directions. Its condition concerns the diffusion of that ratio. A government can therefore have no reason to change safe debt while still wanting a different risky exposure, or want to borrow intertemporally even when the available risky claim provides no useful insurance.

This separation also clarifies the role of expected returns. The premium $\sigma_R'\lambda_I$ tells the government what the market pays for bearing the claim's risk. It enters ordinary return demand. Insurance depends on how the same payoff covaries with fiscal marginal value. Reporting only the total position adds these two motives together and can conceal both of them. The portfolio decomposition in Section 4 makes the distinction explicit.

3.7 Valuation along the traded direction

For a fixed continuation value, the part of the Hamiltonian that changes with s is

$$\mathcal{H}^s(s) = s\beta_R V_N + sD[V] + \frac{1}{2}s^2\|\sigma_R\|^2 V_{NN}, \quad D[V] = \sigma_z\sigma_{Rz}V_{Nz} + \sigma_a\sigma_{Ra}V_{Na}. \tag{3.12}$$

The first term is the compensated market return. The second is the covariance of the claim with the fiscal marginal value inherited from the two primitive states. The last term is curvature. Differentiating and substituting (3.7) and (3.11) gives the cancellation that matters:

$$\begin{aligned}\frac{\mathcal{H}_s(s)}{V_N} &= \beta_R + \frac{D[V] + s\|\sigma_R\|^2 V_{NN}}{V_N} \\ &= \sigma_{Rz}[\lambda_{Iz} - \lambda_{Gz}(s)] + \sigma_{Ra}[\lambda_{Ia} - \lambda_{Ga}(s)].\end{aligned} \tag{3.13}$$

Only after displaying the two components is it convenient to write this as

$$\frac{\mathcal{H}_s(s)}{V_N} = \sigma_R'[\lambda_I - \lambda_G(s)]. \tag{3.14}$$

Proposition 3.1 (Traded-direction fiscal valuation)**.** *On a smooth Brownian continuation with $V_N > 0 > V_{NN}$, a nonzero claim loading σ_R, and the maintained public-budget and market-pricing conventions, equation* (3.14) *is an exact Hamiltonian identity. If s^* is an unconstrained interior maximizer, then*

$$\sigma_R'[\lambda_I - \lambda_G(s^*)] = 0. \tag{3.15}$$

It is a necessary condition for that smooth interior candidate, not an existence, sufficiency, uniqueness, or complete-markets result.

The condition is an alignment statement. A public claim is attractive at the margin when its payoff has positive projection on the innovation in Θ; equivalently, the instantaneous covariance of dR with $d\Theta/\Theta$ is positive. One familiar example is worker downside: wages and worker resources fall more than market-pricing consumption, so a claim that pays in that state transfers valuable resources to the fiscal system. A second, economically distinct case is unequal upside: workers gain, but investors gain more, so relative worker marginal value still rises. The result does not require a literal worker loss. Conversely, if worker consumption moves proportionately with the market-pricing consumption, there is no relative-valuation innovation along the claim.

Portfolio bounds change equality into one-sided conditions. With a signed holding interval $[\underline{s}, \overline{s}]$ and the stated concavity,

$$\begin{aligned} s^* = \underline{s} \quad &\Longrightarrow \quad \mathcal{H}_s(s^*) \leq 0, \\ \underline{s} < s^* < \overline{s} \quad &\Longrightarrow \quad \mathcal{H}_s(s^*) = 0, \\ s^* = \overline{s} \quad &\Longrightarrow \quad \mathcal{H}_s(s^*) \geq 0. \end{aligned} \tag{3.16}$$

The nonzero projected wedge is an outward-pointing marginal derivative at a bound: at the lower bound the value of moving outward is $-\mathcal{H}_s$, while at the upper bound it is $\mathcal{H}_s$. It is not the same thing as an unspanned state direction, and it is not a measure of debt capacity. The separately compiled theory appendix supplies the full derivation and records the required smoothness, admissibility, and sign conventions.

4 What the public payoff menu can insure

The preceding condition says when the government stops changing its position along a given payoff. It does not say that the fiscal and market valuations have become equal state by state. With two Brownian shocks and one claim, the public balance sheet can choose one exposure direction. The remaining direction is a missing payoff, not a failure to take a sufficiently large position in the existing claim.

4.1 Projection before welfare

Let J denote the Brownian present value of future worker wages and net fiscal resources on the deterministic continuation used for the small-risk expansion. We now impose the specialized aligned small-risk closure used for the leading-order portfolio decomposition: $\widehat{\sigma}_R = \widehat{\lambda}_I$, with all Brownian loadings scaled by ε. Its two exposure components are

$$\zeta_J = (\zeta_{Jz}, \zeta_{Ja})' = (\widehat{\sigma}_z J_z, \widehat{\sigma}_a J_a)'. \tag{4.1}$$

Here hats denote quantities after the common small-risk scaling; J remains a worker-resource value, not Treasury collateral. The scalar amount of that exposure replicated by one unit of the payoff is

$$h_J = \frac{\zeta_{Jz}\widehat{\sigma}_{Rz} + \zeta_{Ja}\widehat{\sigma}_{Ra}}{\widehat{\sigma}_{Rz}^2 + \widehat{\sigma}_{Ra}^2}. \tag{4.2}$$

Subtracting the reachable part gives component-by-component residuals

$$\zeta_{Jz}^{\perp} = \zeta_{Jz} - h_J\widehat{\sigma}_{Rz}, \qquad \zeta_{Ja}^{\perp} = \zeta_{Ja} - h_J\widehat{\sigma}_{Ra}. \tag{4.3}$$

Multiplying the first residual by σ_{Rz}, the second by σ_{Ra}, and adding yields

$$\widehat{\sigma}_{Rz}\zeta_{Jz}^{\perp} + \widehat{\sigma}_{Ra}\zeta_{Ja}^{\perp} = 0. \tag{4.4}$$

Thus the residual is orthogonal to the existing payoff. In compact notation, $\zeta_J = h_J\widehat{\sigma}_R + \zeta_J^{\perp}$ and $\zeta_J^{\perp} = (I - P_R)\zeta_J$, where P_R projects onto the line spanned by $\widehat{\sigma}_R$.

Proposition 4.1 (Payoff span and residual exposure)**.** *For a nonzero Brownian payoff*

Table 1: How to read the relative-valuation condition

State experiment	Worker resources relative to investors	Claim payoff	Fiscal-hedge implication
Worker downside	Workers lose more	Positive	A long claim position supplies insurance.
Unequal shared upside	Both gain, investors gain more	Positive	A long position shares an upside that workers value relatively highly.
Common exposure	Both move proportionately	Any	No relative-valuation motive along that payoff.
Claim-neutral displacement	Worker or fiscal resources move	Zero on impact	The existing claim cannot insure the impact movement.

loading, the decomposition in (4.2)–(4.3) *is unique. A scalar public position can offset the marketed component by taking the opposite exposure,* $-h_J\widehat{\sigma}_R$*, but cannot alter* $\zeta_J^{\perp}$*. The local fiscal exposure is fully spanned if and only if* $\zeta_J^{\perp} = 0$*. A second payoff completes this two-dimensional local span only if its loading is linearly independent of* $\widehat{\sigma}_R$*.*

This is exact linear geometry under the named local exposure mapping. It is not a claim that workers obtain private asset access, that all equilibrium constraints are relaxed, or that the resulting allocation is first best. An asset menu can be rank-complete and still fail to finance a desired transfer or violate tax, solvency, or gross-position constraints.

Table 1 translates the projection into four state experiments. The table is intentionally qualitative: the paper does not replace a missing computed exposure with a schematic numerical figure.

4.2 Return demand and the fiscal hedge

The projection becomes a portfolio formula after separating deterministic resources from small Brownian risk. Let $X = N + J$ be comprehensive worker-planner resources and scale return loadings and market prices of risk by ε. On the stated unconstrained branch, the leading position is

$$s^0 = \underbrace{X}_{\text{compensated-return demand}} \underbrace{-h_J}_{\text{fiscal-resource hedge}} . \tag{4.5}$$

The aligned normalization gives $s^0 = X - h_J$. A general external claim has a different return coefficient and requires a separately derived generic-claim representation. The identity separates two reasons for a gross holding. A positive premium can generate return demand even when the claim has little insurance value. A large hedge can offset most of that demand and leave a small net position even when the claim is economically important.

The formula does not establish feasible leverage. In particular, J contains worker wages because it measures resources relevant to worker welfare, while the fiscal capacity of a government issuing debt depends on the distinct tax-backed viability problem. A desired residual safe borrowing position is therefore not a natural debt limit.

4.3 The welfare statement is local and separate

The exact projection proposition should not be mistaken for an exact welfare ranking. The bridge from valuation to welfare is the small-risk fiscal-resource problem. Along the common deterministic continuation, innovations move the present value J of worker and tax resources. Projecting that exposure onto the marketed return is the same local operation as projecting the diffusion of the fiscal SDF relative to the market SDF: the marketed component can be offset by the inherited claim, whereas the orthogonal component remains in marginal worker resources. Squaring that residual and scaling it by the curvature of log continuation value produces the second-order loss below. This does not turn the exact SDF condition into a global welfare theorem. Conditional on a regular small-risk expansion, a common feasible deterministic path, and slack constraints, the additional local value of a completing payoff takes the order-ε^2 form

$$V^{\text{complete}} - V^{\text{one}} = \varepsilon^2 \int_0^\infty e^{-\rho_W t} \frac{\|\zeta_{J,t}^{\perp}\|^2}{2\rho_W X_t^2}\, dt + O(\varepsilon^4). \tag{4.6}$$

The one-claim gain is likewise a completed-square expression involving the leading position. These formulas say why the residual is the relevant missing-insurance object. They do not supply a finite-risk welfare magnitude, a global policy ranking, or a certified error bound. The proof requires optimizer selection, regularity, transversality, and a domain that stays away from active constraints.

4.4 No claim, one claim, and completed public span

The economic value of span is defined by an asset-menu comparison, not by the level of the chosen holding. Begin with the same inherited economy and remove access to the risky claim. The government can still use safe finance and the capital-income tax, but neither instrument delivers a contemporaneous payoff across the two innovations. Worker resources therefore inherit the full local exposure ζ_J.

Next give the government the maintained external claim. It can choose the scalar h_J and remove the projection of ζ_J onto the claim loading. This comparison holds the technology, market prices, tax instrument, and inherited state fixed. Its gain is the value of the particular payoff direction, not the value of public balance sheets in general. If the claim is poorly aligned with fiscal exposure, the gain can be small even when the claim has a large premium or a familiar automation label.

Finally, add a second claim whose loading is linearly independent of the first. To isolate insurance, price the added claim under the same world discount factor and normalize its expected excess return to zero. The added security then contributes no ordinary

Table 2: Asset-menu comparisons

Public menu	Reachable state transfer	What changes	What does not follow
Safe finance only	Timing of resources	Intertemporal fiscal exposure	No contemporaneous innovation is insured.
One risky claim	Projection on σ_R	Marketed part of fiscal exposure and return opportunity	The orthogonal residual need not fall to zero.
Two independent claims	Both local innovation directions	Public payoff span for the named exposure	Households do not thereby obtain complete markets; finance and constraints may still bind.

compensated-return opportunity. Its value comes from reaching the residual $\zeta_J^\perp$. This zero-premium completion experiment is the cleanest way to show that payoff span, rather than average return, is the object at issue.

The final row is deliberately called completed *public payoff span*. It is not a complete-markets allocation. Workers remain hand to mouth. Domestic owners remain segmented from international asset markets. The government may lack the wealth or borrowing capacity needed to take the replicating positions. Capital taxation still distorts accumulation, and the worker-only welfare criterion still differs from a first best defined over all domestic households. Rank removes one friction and leaves the others intact.

The comparison also clarifies why a large public fund and a well-spanned public balance sheet are different objects. Scaling a fund raises the quantity of every payoff it can hold. It does not create a missing payoff direction. Adding a new claim can improve span even when its unconditional premium is zero and its average holding is modest. Size is a financing object; composition is a state-contingency object.

4.5 A local numerical diagnostic

One exploratory Brownian calculation is useful because it makes a potential misreading of (4.5) concrete. At one checked productivity node under illustrative inputs, compensated return demand is about 12.70 and the fiscal hedge is about -12.25, leaving a leading net position of roughly 0.45. At checked nodes the maintained claim spans about 99.8% of squared raw fiscal exposure. These are diagnostic values from a local calculation, not calibrated policy numbers. They neither show that the unspanned residual is generally small nor justify a welfare magnitude, a debt-financing conclusion, or a recommendation about public equity. The relevant lesson is narrower: a near-zero net holding can conceal large and offsetting return and insurance components.

The exact-in-deterministic-state corrections to tax speed and the risky position are kept in the computational appendix. They are method diagnostics, not evidence that risk lowers taxes or public insurance. The theory appendix gives the derivation of the projection and local welfare formula, together with a trace of the assumptions under which those equations may be used.

4.6 Three comparisons that keep the geometry economic

The projection is easiest to misunderstand when the shock is described only as "automation." Three local comparisons prevent that shortcut. First, an aligned innovation lowers worker resources relative to the market investor and makes the maintained claim pay. Its projection is large because the claim and the fiscal exposure point in similar directions. This is the familiar downside-insurance case, but it is not the only way in which Θ can rise.

Second, consider a claim-neutral innovation. At impact it is constructed so that the external claim has zero payoff, while wages and the fiscal-resource value can still move. In the primitive two-shock representation this means choosing a local direction orthogonal to the claim loading. A claim holding has no impact ability to offset this movement. Different shock persistence can subsequently make even an initially output-neutral or claim-neutral direction non-neutral; these thought experiments identify an impact mechanism, not a complete transition counterfactual.

Third, add a hypothetical second public payoff with loading proportional to $\zeta_J^{\perp}$. Price it under the same world SDF and choose the normalization so that it has no local excess-return premium. Demand for this payoff is then a pure missing-direction hedge in the local benchmark. The thought experiment identifies what one extra payoff direction would do. It does not say that the security exists, that the government can hold the required quantity, or that adding it changes neither prices nor institutions.

These comparisons distinguish two reasons a public balance sheet may fail to provide insurance. If the existing holding is at a bound, the government lacks room along a marketed direction. If $\zeta_J^{\perp} \neq 0$, no increase in that same holding reaches the residual direction. Taxes can change future J and hence the exposure to be hedged, but a predictable flow tax supplies no contemporaneous Brownian payoff atom. Safe finance moves resources across dates. None of these statements determines which instrument is quantitatively more valuable on a constrained transition.

4.7 Fiscal wealth connects tax policy to the portfolio

Suppress instantaneous Brownian risk and freeze the common exogenous state at the reference point. Let $\mathcal{F}(K,\tau)$ be worker wages plus net rental-tax resources. The deterministic fiscal-resource problem is

$$J(K,\tau) = \sup_{\{\nu_t\}} \int_0^{\infty} e^{-\rho_W t} \left[\mathcal{F}(K_t,\tau_t) - \frac{Y(K_t)}{2\kappa_\tau} \nu_t^2 \right] dt, \quad \dot{K} = K g(K,\tau), \quad \dot{\tau} = \nu. \tag{4.7}$$

Wages enter J because they change worker welfare; they do not become Treasury collateral. The interior equation is

$$\rho_W J = \mathcal{F} + KgJ_K + \frac{\kappa_\tau}{2Y} J_\tau^2, \qquad \nu = \frac{\kappa_\tau}{Y} J_\tau. \tag{4.8}$$

With $X = N + J$, the selected feasible deterministic interior characteristic has the log consumption rule $c^W = \rho_W X$. This is a separation of worker resources from financial net worth, not a solvency theorem. If the costate has the required terminal limit, tax adjustment obeys

$$\nu_t = \frac{\kappa_\tau}{Y_t} \int_t^\infty e^{-\rho_W(u-t)} \mathcal{B}_u [1 - J_K(S_u)] du. \tag{4.9}$$

The tax base supplies direct fiscal resources; J_K is the opposing fiscal value of preserving capital. The formula explains why current tax-base news cannot by itself sign a tax response. It also separates the roles of the instruments: taxation changes future fiscal wealth, safe finance moves resources across dates, and a risky claim changes exposure at the current innovation.

4.8 Reading the relative-SDF condition one component at a time

The compact condition in (3.14) can obscure the economic comparison. The productivity component of the return contributes $\sigma_{Rz}\lambda_{Iz}$ to its market premium. The same component contributes $\sigma_{Rz}\lambda_{Gz}$ to the fiscal valuation of that payoff. Their difference is the contribution of the productivity innovation to the value of raising the public holding. The automation component makes the analogous comparison with $\sigma_{Ra}\lambda_{Ia}$ and $\sigma_{Ra}\lambda_{Ga}$. Adding the two components does not introduce a new assumption: it simply values the same traded return from the two sides of the economy.

The quotient $\Theta = m_G/m_I$ puts this comparison in a form that does not depend on which of the two SDFs is used as the starting point. Its innovation is

$$\left.\frac{d\Theta_t}{\Theta_t}\right|_{dB} = (\lambda_{Iz} - \lambda_{Gz})dB_t^z + (\lambda_{Ia} - \lambda_{Ga})dB_t^a. \tag{4.10}$$

Multiplying this expression by the return innovation in (3.5) and taking the instantaneous covariance gives

$$\frac{\mathrm{Cov}_t(dR_t, d\Theta_t/\Theta_t)}{dt} = \sigma_{Rz}(\lambda_{Iz} - \lambda_{Gz}) + \sigma_{Ra}(\lambda_{Ia} - \lambda_{Ga}). \tag{4.11}$$

This is exactly the right-hand side of the portfolio condition. A marginal long position is valuable when the claim pays in innovations that raise the relative fiscal value of a public dollar. The expression is a covariance of innovations, not a claim that the level of inequality, the expected return, or the asset's sector label is the relevant sufficient statistic.

Three cases make the condition concrete. In a *worker-downside* state, worker resources fall while market-pricing consumption falls less or rises; Θ rises and a claim that pays in that state is a fiscal hedge. In an *unequal-upside* state, workers and investors both gain but investors gain more. Relative worker marginal value again rises, so the same claim can share an upside without being an insurance claim against an absolute worker loss. In a *worker-outperformance* state, workers do better relative to investors; the fiscal hedge points away from a claim that pays there. A positive expected premium may still make the total holding long. The first two cases support the same sign of the relative-valuation projection, but they are economically different stories; neither should be elevated into a universal interpretation of automation.

4.9 A scalar projection example

Take a two-coordinate fiscal exposure (a, b) and a one-claim payoff (r_z, r_x). The scalar $h = (ar_z + br_x)/(r_z^2 + r_x^2)$ is chosen so that $(a - hr_z)r_z + (b - hr_x)r_x = 0$. The claim can offset the first vector component only in the fixed ratio $r_z : r_x$; it cannot choose a separate coefficient for a and b. If (a, b) is exactly a multiple of (r_z, r_x), the residual is zero. If it is not, the residual remains even if the government is free to take an arbitrarily large position in the existing claim.

This scalar example is an exact statement about a payoff line. The local welfare formula adds further structure: a common deterministic path, a small-risk scaling, and a feasible comparison between asset menus. The squared residual enters welfare only after those assumptions turn a local quadratic into the order-ε^2 expression in (4.6). Exact geometry therefore tells the reader what risk the menu cannot reach; it does not tell the reader how large the finite-risk welfare cost is.

4.10 Completion, capacity, and a guide to the claims

With several claims, completion means only that their loading columns span the local exposure vector. The Gram-matrix formula in (4.13) constructs the least-squares exposure; it does not finance the resulting positions. If a holding limit binds, the KKT condition says the marginal value of moving farther along an existing column is nonzero. That is a capacity problem. If the residual is orthogonal to every available column, relaxing the same limit does not reach it. That is a missing-payoff problem. A constrained economy may face both at once.

The reader can use the propositions accordingly. Proposition 3.1 is an exact necessary condition for a smooth interior candidate; its strongest qualification is that it supplies neither existence nor sufficiency. Proposition 4.1 is exact linear geometry for the named local payoff mapping; its strongest qualification is that payoff rank is not household market completion or implementation. Equation (4.5) is a leading small-risk decomposition; its strongest qualification is that desired safe borrowing is not a debt-capacity result. Finally, (4.6) is an order-ε^2 welfare comparison whose use depends on the common feasible path and the regular expansion. These qualifications are part of the economic meaning of the results, rather than after-the-fact disclaimers.

4.11 Several public claims

The one-claim result is not tied to two shocks. If a menu of q claims has loading matrix $\Sigma_R = [\sigma_{R1}, \ldots, \sigma_{Rq}]$, a vector of holdings can affect only the column space of Σ_R. The multi-claim version of the interior condition is

$$\Sigma'_R[\lambda_I - \lambda_G] = 0. \tag{4.12}$$

When the Gram matrix $\Sigma'_R\Sigma_R$ is invertible, the local fiscal-exposure projection is

$$P_\Sigma \zeta_J = \Sigma_R(\Sigma'_R\Sigma_R)^{-1}\Sigma'_R\zeta_J, \qquad \zeta_J^\perp = (I - P_\Sigma)\zeta_J. \tag{4.13}$$

The formula records a mathematical span, not a feasible portfolio. Rank can be high while transfer floors, tax bounds, borrowing restrictions, equilibrium recovery, or the continuation-solvency set still prevent the corresponding allocation. Conversely, a low-rank menu may be sufficient for the particular fiscal exposure at a particular state when that exposure happens to lie in its column space.

The residual also has a useful accounting property. Orthogonality implies the Pythagorean decomposition

$$\|\zeta_J\|^2 = \|P_\Sigma\zeta_J\|^2 + \|\zeta_J^\perp\|^2. \tag{4.14}$$

The ratio of the first term to the left-hand side is a local descriptive measure of spanned squared exposure. It is not a general measure of welfare insurance: welfare also depends on the level of X, persistence, the continuation policy, and whether the positions needed to implement the projection are feasible.

4.12 Why the relative SDF is the right covariance object

The model contains two different questions that are often merged. One is whether the claim has a positive market premium. The other is whether it pays when a public dollar has high marginal value for workers. The first question is answered by $\sigma_R' \lambda_I$; the second compares that object with $\sigma_R' \lambda_G$. Their difference is not an additional preference for risk. It is the wedge between a marginal investor's valuation and the fiscal value of the same payoff.

This distinction is visible in a log-consumption benchmark. If the external SDF has the usual consumption representation, λ_I is the diffusion of market-pricing consumption. The worker-consumption FOC, $c = 1/V_N$, makes λ_G the diffusion of worker consumption on the maintained interior branch. The interior condition therefore sets the claim covariance with the diffusion of $\log(c/C^I)$ to zero. It does not set the variance of c to zero, set consumption equal across agents, or eliminate variation in Θ in directions outside the payoff span.

The relative formulation also gives two useful negative controls. If worker and market-pricing consumption have the same innovation loading along the claim, the insurance component is zero even if both groups face substantial risk. If a claim pays strongly in states in which workers do well relative to investors, the fiscal hedge points short even when the claim earns a positive premium. Gross demand can remain positive because return demand and fiscal hedging are distinct terms in (4.5). A holding level alone therefore cannot reveal the direction or magnitude of worker insurance.

4.13 What taxation changes, and what it cannot change on impact

The payoff-span result does not declare taxation irrelevant. A source tax changes the distribution of rental income and changes capital accumulation, which changes the future wage and tax components of J. It can therefore change both the level and the exposure of fiscal resources that the public portfolio seeks to hedge. But the tax rate and installed capital are inherited at a Brownian innovation. A predictable rental-flow tax has no contemporaneous Brownian payoff atom. It cannot independently choose the impact-state transfer that a risky holding supplies.

This timing explains why taxes and risky claims are neither automatic substitutes nor automatic complements. A claim can reduce pressure to use a distortionary tax by delivering resources in a state in which workers need transfers. Alternatively, a tax change can preserve or erode the fiscal-resource base against which portfolio insurance is valued. The sign of this interaction depends on the continuation path of capital, taxes, and constraints. The payoff projection by itself does not sign it. Section 6 returns to the deterministic tax condition; the present sections only establish the state-contingent restriction faced by the portfolio.

5 Capital taxation and transition dynamics

The payoff-span result says which state-contingent transfer a public claim can buy. It does not say that the transfer can be financed, or that the tax system can reproduce an untraded payoff. This section makes that implementation distinction concrete. Safe positions move resources across dates; a risky position moves resources across the directions in its payoff; and the capital-income tax changes both current distribution and the future capital stock. These are different instruments even when all three enter the same public budget.

5.1 Three instruments, three dates

Consider an automation innovation at date t. The external risky holding s_t is inherited immediately before the innovation and therefore delivers its contemporaneous return payoff. Installed capital K_t and the tax rate τ_t are also inherited: they cannot jump merely because news arrives. The government can subsequently alter the tax rate only through its speed $\nu_t = \dot{\tau}_t$, and domestic owners alter capital gradually. Thus an asset can insure an impact state that a slowly moving tax cannot reach; a tax can alter future rental income and accumulation in a way that an asset payoff cannot; and safe finance can transfer a resource between dates without creating a new Brownian payoff direction.

This timing is also why a dollar holding is not a portfolio share and why a desirable hedge is not a debt limit. A signed external position can be financed in part with safe borrowing, but it remains subject to the public budget, transfers, tax bounds, and continuation solvency. Conversely, issuing a safe claim cannot span the automation component left orthogonal to the risky payoff.

The chronology is worth keeping in view. The inherited risky position pays within the innovation window. Capital and the tax rate are predetermined at that instant. Worker resources, taxes, and capital then adjust along the continuation. The impulse responses below show this sequence using output produced by the local solution rather than a schematic timing diagram.

5.2 A forward-looking tax condition

The deterministic interior benchmark isolates the relevant tax trade-off. Let $\mathcal{B}_u$ be signed net rental income, and let $J_K(S_u)$ be the marginal value, for worker resources, of an additional unit of installed capital along the selected continuation. On the deterministic characteristic, the tax speed satisfies

$$\nu_t = \frac{\kappa_\tau}{Y_t} \int_t^\infty D_{t,u} \mathcal{B}_u \left[1 - J_K(S_u)\right] du. \tag{5.1}$$

Here κ_τ is the adjustment-cost scale and $D_{t,u}$ is the discount factor from t to u implied by the local safe-rate path. The display makes clear why the current tax base cannot sign the current tax response. Raising the tax is valuable when future taxable rental income is valuable; it is costly when the capital displaced by the higher future wedge has high fiscal value. Persistence, capital accumulation, and the terminal condition all enter the balance.

Equation (5.1) is a conditional deterministic characterization, not an unconditional tax rule. It requires the stated smooth, interior continuation and the transversality condition selecting it. If a transfer floor, tax bound, portfolio restriction, specialization boundary, or solvency constraint binds, the equality is replaced by the corresponding KKT or state-constraint condition. In particular, neither the rank of the traded payoff menu nor a pointwise correction to ν_t supplies a sign for the tax level τ_t.

5.3 From impact to propagation

The local transition supplies the economic sequence behind this condition. An automation innovation can shift income from wages to capital income while capital is fixed. The initial wage effect is therefore distinct from the later effect of induced accumulation. Subsequently, the tax path changes the owner's after-tax return,

$$\frac{\dot{K}_t}{K_t} = (1 - \tau_t)(R_t^K - \delta) - \rho_K, \tag{5.2}$$

which changes wages, output, rental income, and hence the tax base. The principal opposing force is immediate redistribution versus the loss of future worker-relevant resources through slower capital accumulation. A persistent displacement shock makes the forward-looking term particularly important; a short-lived shock can leave little time for the capital channel to operate.

The local quadratic representation is useful precisely because it keeps this sequence visible: inherited capital and the tax rate determine impact; tax adjustment and capital accumulation propagate the disturbance; and the stable continuation supplies the limit. It is not a claim that a linear feedback is globally optimal. The strict local LQ benchmark characterizes conditional-mean dynamics around a named deterministic interior anchor. Leading portfolio and welfare effects are separate small-risk objects; an order-two coefficient ν^2 is a correction to tax *speed*, not a plotted tax path. The computational appendix explains the approximation ladder and the evidence needed before any finite-risk transition comparison is reported.

5.4 The one-half benchmark

The deterministic steady state gives a useful benchmark for the tax–capital trade-off. Suppose workers and owners share the discount rate ρ and the path converges to an interior point with a positive rental-income tax base and slack constraints. Private capital stationarity requires

$$(1-\bar{\tau})(\bar{R}^K-\delta)=\rho, \tag{5.3}$$

while the fiscal costate condition requires

$$\bar{\tau}(\bar{R}^K-\delta)=\rho. \tag{5.4}$$

Taken together, the two return conditions give the conditional benchmark

$$\bar{\tau}=\frac{1}{2}, \qquad \bar{R}^K-\delta=2\rho. \tag{5.5}$$

The number one half is not a primitive target. It divides the net rental return so that private accumulation and fiscal valuation each receive the common return ρ. The argument is conditional on convergence to this interior branch and does not establish that the global Ramsey path reaches it. If workers and owners discount at different rates, both the transition and the fiscal costate change. The simple two-equation calculation no longer supplies a general closed form for the limiting tax.

This benchmark is consistent with Reinhorn (2019)'s reconciliation of the long-run capital-tax literature. The familiar zero-tax conclusion uses a modified-golden-rule branch on which saving responds to the after-tax return. With logarithmic preferences, income and substitution effects can instead offset so that saving is locally insensitive to that return. The private block used here lies in that exceptional class. Reinhorn's result explains why a positive limiting capital tax is not automatically a contradiction of the zero-tax literature; it does not select the one-half value in this economy. That value comes from the two return conditions above, together with the worker-only Ramsey objective and the maintained private saving rule.

Nor is (5.5) a Laffer-curve calculation. In the constant-share Cobb–Douglas revenue benchmark, let α denote the capital elasticity. Under common discounting, the constant tax rate that maximizes stationary rental-tax receipts is

$$\tau_{\text{Laffer}} = \frac{(1-\alpha)(\rho+\delta)}{\rho+(1-\alpha)\delta}, \tag{5.6}$$

which equals one half only on a parameter knife edge. Revenue maximization asks how a permanent tax changes the stationary base. Ramsey policy also values the wages and future fiscal capacity supplied by the marginal unit of capital.

5.5 Patience, capital deepening, and wages

Automation affects wages through a race between displacement and accumulation. Holding installed capital fixed, a rise in the automation state lowers labour's effective share and can reduce the wage even as it raises output and the rental return. The higher return then induces owners to accumulate. More patient owners consume a smaller part of their wealth and rebuild the capital stock more strongly after the same innovation. That makes a later wage recovery more likely. More patient workers, represented by a lower government discount rate, also place greater weight on this future wage recovery relative to an immediate transfer financed by capital taxation.

The qualitative trade-off is therefore not "wages versus growth." It is current worker resources versus the wage and tax services of capital along the entire transition. A tax can soften the initial distributional loss but weaken the accumulation that would otherwise restore wages. An inherited claim can improve resources on impact without directly lowering the domestic after-tax return. Whether the two instruments are substitutes or complements depends on how the claim payoff, tax base, and capital response line up over time.

5.6 An exact fixed-tax transition

A closed-form benchmark shows how displacement and accumulation can reverse the sign of the wage response. Hold productivity, automation, and the tax rate fixed after an initial change, and write production as

$$Y_t = \mathsf{A} K_t^{\alpha}, \qquad 0 < \alpha < 1. \tag{5.7}$$

The owners' accumulation equation becomes

$$\dot{K}_t = (1-\tau)\alpha \mathsf{A} K_t^{\alpha} - [\rho_K + (1-\tau)\delta]K_t. \tag{5.8}$$

Define

$$\Gamma_\tau = \rho_K + (1-\tau)\delta, \qquad K^* = \left[\frac{(1-\tau)\alpha\mathsf{A}}{\Gamma_\tau}\right]^{1/(1-\alpha)}. \tag{5.9}$$

The power transformation $u_t = (K_t/K^*)^{1-\alpha}$ turns the nonlinear transition into

$$\dot{u}_t = (1-\alpha)\Gamma_\tau(1-u_t). \tag{5.10}$$

Thus capital approaches its fixed-tax steady state at a rate governed by depreciation, owner impatience, the tax wedge, and the curvature of production. This exact benchmark is more informative than reading the sign of an eigenvalue: it states which economic parameters determine the speed at which capital can offset the initial wage movement.

Suppose the automation innovation raises the scale term A while reducing the labour share embedded in the wage schedule. At impact, K_0 is inherited, so the wage responds only through scale and displacement. Along the transition, K_t rises toward the new K^* when the after-tax marginal product exceeds owner impatience. The wage can therefore fall on impact and rise in the long run. A sufficiently patient owner makes the capital response larger and faster; a high tax or high depreciation weakens it. Nothing in the exact solution guarantees reversal, because the lasting displacement of labour may dominate the scale and capital-deepening effects.

The benchmark also separates the effect of tax flexibility from the effect of a risky claim. A lower tax raises the private return and speeds accumulation, but it gives up current fiscal resources. A claim payoff changes public net worth at the innovation and does not directly alter equation (5.8). The Ramsey plan chooses the tax path by valuing both current transfers and the future wage services of capital. The portfolio condition chooses exposure by valuing the state in which public resources arrive. Treating the claim as "another way to tax capital" would merge these two distinct margins.

5.7 Why local dynamics can overshoot

The fixed-tax transition is monotone in the transformed capital state, but the Ramsey transition need not be monotone in taxes, capital, or worker consumption. Productivity and automation are persistent forcing processes; the tax rate responds gradually; and capital carries its own endogenous propagation. An observed impulse response is a mixture of these components. Even when the intrinsic capital–tax roots are real, their eigenvectors can enter an individual variable with opposite signs and generate overshooting. Complex roots classify the local invariant modes, but do not by themselves imply an economically meaningful cycle.

This distinction matters for interpretation of numerical paths. A tax rate that first falls and later rises is not evidence that the government changes objectives. It may reflect an initial desire to preserve accumulation followed by the later value of the expanded tax base. Likewise, a worker-consumption path can cross its steady-state level because the wage, transfer, and public-asset channels peak at different dates. The local solution is useful when it decomposes these forces; a root classification alone is not a paper result.

5.8 Inherited capital and the institution behind tax flexibility

The maintained Brownian continuation starts from inherited productivity, automation, capital, public net worth, and tax rate. Capital and the tax rate are continuous. The government cannot impose an instantaneous levy on installed capital, but it can alter future flow taxes while that capital remains in place. When the net-rental base is positive, this creates a transition quasi-rent.

That statement is institutional, not universal. A grandfathered tax treatment, protected owner claim, or promised continuation value would alter the inherited fiscal state, its shadow value, and the tax problem. Nor does comparing a frozen and adjustable tax isolate only incumbent protection: it simultaneously changes state-contingent redistribution and the evolution of future fiscal resources. These qualifications are central because the maintained objective is worker continuation welfare, not an aggregate domestic Pareto criterion.

5.9 Fiscal value is not fiscal capacity

It is helpful to separate three objects. Public financial net worth N_t is a balance sheet stock. Fiscal resource value J_t values resources relevant to worker consumption. Comprehensive resources $X_t = N_t + J_t$ are useful for the local valuation and portfolio decomposition. Neither J_t nor X_t is pledgeable Treasury collateral: worker wages enter worker welfare but are private and nonpledgeable.

The relevant capacity question is instead whether at least one admissible continuation can satisfy conditional solvency, no-Ponzi, and policy restrictions after every stopping time. Its structural boundary is distinct from both a conservative reference floor used to screen a computation and the outer faces of a numerical box. A path remaining above the reference floor is evidence about that screen, not an estimate of a maximum debt limit. This distinction prevents a familiar but consequential mistake: converting a present value or an unconstrained desired risky position into a claim about feasible borrowing.

5.10 What a transition comparison must hold fixed

In a commitment problem, a transition is a continuation from the same inherited state, not a sequence of fresh planners who rediscover the physical state after every shock. The comparison must name inherited capital, tax rate, public net worth, portfolios, pricing convention, and surviving commitments. Holding output or debt alone fixed is not enough: an inherited position changes both the feasible set and the distribution of rents. To compare no risky access, one claim, and an expanded payoff menu, each continuation must start from the same inherited state and differ only by an instrument restriction. To compare a frozen and adjustable tax, retain τ_0 but solve the frozen economy as its own characteristic; setting the adjustable solution's ν to zero after solving does not construct the frozen-tax economy. The point is economic, not procedural: a reported difference must have one intelligible cause.

5.11 Local dynamics as a mechanism laboratory

The local system is useful before it is quantitatively decisive. Suppose automation moves while inherited capital, the tax rate, and the balance sheet cannot. Worker wages may fall at impact even when output or the rental base rises. The tax cannot repair that impact instantly; an inherited risky payoff can. Thereafter the tax path changes the owner's after-tax return, capital changes wages and rental income, and the tax base feeds back into the fiscal problem. A persistent disturbance makes the forward-looking capital term important; a short-lived disturbance leaves little time for it to operate. Likewise, a risky claim has no insurance role in a direction orthogonal to its payoff, even if it has positive expected return. These are timing and direction statements, not a global policy comparison.

5.12 A practical taxonomy of fiscal limits

Accounting feasibility, institutional bounds, and continuation solvency are distinct. The first requires realized public flows to add up. The second limits taxes, transfers, or positions. The third asks whether an admissible continuation exists after future histories. A positive fiscal present value can coexist with failure of either of the latter restrictions. A conservative reference floor supplies a screen and a place to measure slack; it does not determine a maximum debt limit, reveal how the boundary moves with prices, or establish viability elsewhere. If a conclusion changes when the floor moves modestly, the appropriate next object is the boundary problem, not a relabelled floor.

5.13 The local system: three blocks, three economic jobs

The local LQ representation is deliberately organized around the economics rather than around a large matrix. The nonlinear part is a 2×2 Riccati block for the joint capital–tax feedback. It asks how a marginal tax change trades current worker resources against the induced change in the after-tax return to installed capital. Its stabilizing solution selects the local feedback that prevents the capital–tax subsystem from drifting away from the deterministic interior anchor.

Conditional on that block, a Sylvester equation determines the cross effects of the exogenous productivity and automation states on capital and taxes. Economically, it tells us how a shock's persistence is translated into the gradual adjustment of the tax wedge and the capital stock. A discounted Lyapunov equation then accumulates the quadratic welfare cost of those propagating deviations. Discounting makes this value calculation finite; it does not imply that public net worth has a stationary distribution. The safe balance-sheet direction is a level and feasibility object, not one of the capital–tax feedback modes.

This decomposition has two reader-facing implications. First, a Riccati residual is not an economic result; it is evidence that the local capital–tax feedback has been computed consistently. Second, the local transition can be nonmonotone even when the intrinsic modes are real, because an observed path combines forcing from the exogenous states with the endogenous feedback modes. Complex modes, when they arise, classify the local invariant subspace; they do not by themselves prove cycles or sign an individual tax path. The equations and stability conditions are reported in the theory and computational appendices so that the main text can retain the economic sequence: shock, inherited stocks, gradual tax response, capital propagation, and limiting continuation.

5.14 Illustrative local transitions

Figure 6 reports four conditional-mean responses from the development-stage local solution. Each line begins from a named displacement of the same deterministic interior anchor. The upper-left panel shows worker consumption; the upper-right panel shows the tax rate; the lower-left panel shows capital; and the lower-right panel shows public net worth. Productivity and automation innovations have small effects under the illustrative scaling. The output-neutral combination produces a larger rise in worker consumption on impact, followed by capital accumulation and a delayed tax adjustment. The claim-and-rental-base-neutral combination isolates a direction whose impact cannot be repaired by the maintained claim.

Figure 7 holds the two primitive innovation experiments fixed and compares worker-consumption paths with and without access to the maintained external claim. The differences are small in this illustration. Their sign is not the important object: the figure makes the comparison in worker consumption rather than in the gross public position, and the two innovations need not be equally well aligned with the claim payoff.

5.15 A narrow small-risk diagnostic

The available local calculation has one admissible use in the body: it illustrates why a net risky position is not an insurance statistic. At one checked node, a large compensated-return component and a nearly offsetting fiscal hedge leave a much smaller net holding. This is a pointwise mechanism diagnostic under illustrative inputs, not a calibrated portfolio result. It does not show that the unmarketed residual is generally small, nor does it provide a tax path, impulse response, welfare comparison, or finite-risk recommendation.

A cheaper truncated closure gives economically different order-two corrections despite small residuals on the object it shares with the maintained calculation. The difference arises because truncating the state before differentiation omits source derivatives needed for policy recovery. The numerical values and error envelopes for this R3–R4 diagnostic are confined to the Computation and Evidence Appendix. The body retains the methodological conclusion: equation residuals on a shared approximation cannot validate an economic coefficient whose defining derivatives are missing.

6 One arrival, two possible economies

The Brownian economy treats automation as a sequence of small shocks. Some of the questions raised by advanced automation have a different timing. A capability may arrive at an uncertain date and, once it arrives, move the economy permanently into a new production regime. The central public-finance question is then not whether a portfolio can smooth an infinitesimal innovation. It is whether the balance sheet chosen before the arrival delivers resources in the successor economy in which workers need them.

This section develops that question in a small open economy. The first case has one possible labour-free successor and makes the accounting transparent. The second has two possible successors: partial automation, in which labour remains productive, and full automation, in which it does not. The distinction is economically important. Uncertainty is not only about the size of a productivity gain. It is uncertainty over which economy workers will inhabit. The public can trade installed domestic equity before the arrival, but it cannot wait to observe the successor and then choose the position that would have paid in that successor.

6.1 The economy before the arrival

The pre-arrival economy has workers, domestic owners, a government, and foreign investors. It produces with capital and labour. Write its production function as

$$Y_0 = \mathcal{F}_0(K, L), \qquad W_0 = \mathcal{F}_{0,L}(K, L)L, \qquad R_0^K = \mathcal{F}_{0,K}(K, L). \tag{6.1}$$

The notation is intentionally general. The event analysis does not need a particular elasticity of substitution or a particular decomposition of automation into factor augmenting technology and task displacement. It needs a clear accounting point: before the arrival, workers receive the wage bill W_0 and owners receive returns on capital. After a labour-free arrival, the wage bill is zero while the physical stock of capital remains in place.

Installed capital is accumulated by firms. Let ι denote investment expenditure per unit of capital and let $\Phi(\iota) - \delta$ be the net growth rate of the stock. Thus

$$\dot{K} = [\Phi(\iota) - \delta]K. \tag{6.2}$$

Adjustment costs make the distinction between a change in a price and a change in a physical quantity useful. At an arrival, shareholders immediately learn the value of the rental stream generated by the inherited stock. Firms do not immediately install a different stock. Subsequent investment changes K only through equation (6.2).

Table 3 summarizes the timing restriction that carries most of the economics. It prevents two common shortcuts. First, a government cannot replicate an event-contingent transfer merely by announcing a tax rate after it observes the event. The tax acts on subsequent rental flows. Second, a government that wishes to insure an arrival must arrange its relevant exposure before the event, at the price that foreign investors are willing to pay.

Table 3: Timing of the main instruments at an arrival

Object	At the arrival	Thereafter
Installed capital K	Inherited and continuous	Changes through investment
Installed-equity price q	Can revalue immediately	Reflects future rents, taxes, and investment
Public equity position Θ	Delivers its inherited gain or loss	Can be rebalanced after the successor is known
Safe debt B	Has no successor-specific gain	Shifts resources across future dates
Source tax τ^K	Has no tax atom on the revaluation	Alters rental income, investment, and capital prices

6.2 Timing, households, and public accounts before the arrival

Time is continuous. Until the arrival, the economy is in regime 0. An arrival occurs with physical intensities λ_j, where j indexes the possible successor economies. Conditional on arrival, the successor is absorbing. Installed capital cannot jump at that date; its price can. This is the elementary distinction behind the exercise: the physical stock is inherited, whereas the value of its future rents is reassessed when the production regime changes.

Workers do not trade assets. Their consumption is current wages plus the transfer chosen by the government,

$$C_t^W = W_t + T_t, \qquad T_t \geq 0, \tag{6.3}$$

and their date-zero objective is

$$\mathbb{E}_0 \int_0^\infty e^{-\rho_W t} \log C_t^W \, \mathrm{d}t. \tag{6.4}$$

The no-asset assumption is not intended as a literal description of every household. It isolates the public-insurance motive. If workers could buy the same claims in unconstrained private markets, the government would face a different distributional and implementation problem. Here the wage bill is the private resource that is exposed, and the transfer is the public instrument that offsets that exposure.

Domestic owners hold domestic equity and the outside safe asset. With log utility and a homogeneous opportunity set, their consumption is proportional to their financial wealth A_t:

$$C_t^K = \rho_K A_t. \tag{6.5}$$

To see where this rule comes from, let π_t denote the fraction of owner wealth exposed to domestic equity and let r_0^w be the world safe rate. Before an arrival, owner wealth obeys the flow budget

$$\dot{A}_t = \left[r_0^w - \pi_t \sum_j \lambda_j^* G_j \right] A_t - C_t^K. \tag{6.6}$$

At an arrival to successor j, it changes to

$$A_j^+ = A(1 + \pi G_j), \qquad 1 + \pi G_j > 0. \tag{6.7}$$

The inequality is a solvency restriction: an owner cannot choose an equity position that makes wealth negative in a successor economy. With log preferences, the consumption rule in (6.5) follows from the usual homothetic portfolio problem. The private first-order condition selects exposure by trading the physical probability-weighted gain against its world-priced cost. The government takes that private problem as part of equilibrium; it does not allocate owner wealth directly.

Foreign investors hold the residual supply at exogenous world prices. This small-open closure separates the public insurance question from a domestic asset-pricing problem. It also means that public purchases of domestic equity are paid for at the world price; they are not a free claim on domestic capital.

Let q denote the price per unit of installed capital immediately before the arrival, and let $q_j(K)$ be the price in successor j when the inherited capital stock is K. The government holds domestic equity with gross dollar market value Θ; Θ/q is the corresponding quantity of claims. It issues safe debt $B \geq 0$, so its net financial wealth is $F = \Theta - B$. The source tax is a predictable levy on rental flows,

$$\mathrm{d}\mathcal{T}_t^K = \tau_t^K R_0^K(K_t)K_t \,\mathrm{d}t, \qquad 0 \leq \tau_t^K \leq 1. \tag{6.8}$$

It does not tax the revaluation of installed capital at the arrival. A successor tax can change future rents and therefore the successor price, but that is a continuation choice. It is not a second inherited event payoff.

The physical intensities govern probabilities and welfare. World prices instead use risk-neutral intensities λ_j^*. The two need not agree. Letting M^w be the world stochastic discount factor, their relationship is

$$\lambda_j^* = \lambda_j \frac{M_j^{w,+}}{M_0^{w,-}}. \tag{6.9}$$

This wedge is the discrete-event analogue of the gap between the worker's valuation of fiscal resources and the valuation embedded in traded prices.

It is useful to work with public wealth net of the value of installed capital. Define

$$e \equiv F - qK, \qquad \psi \equiv \Theta - qK. \tag{6.10}$$

The first object is a reduced public balance sheet; the second is the public exposure to the revaluation of installed equity. They recover the gross positions through $B = \Theta - F = \psi - e$. Thus the restriction $B \geq 0$ is not innocuous: it requires

$$\psi \geq e. \tag{6.11}$$

It rules out financing a long safe-asset position by negative public debt. A desired insurance position is not automatically a feasible one.

6.3 A labour-free successor with AK production

Start with the polar case in which the arrival eliminates labour income. Output in the successor is produced directly from installed capital,

$$Y_t = AK_t, \qquad R_t^K = A, \tag{6.12}$$

where A is the productivity of capital. Investment per unit of capital is ι_t, and the capital law is

$$\frac{\dot{K}_t}{K_t} = \Phi(\iota_t) - \delta. \tag{6.13}$$

There is no wage in this successor. Workers can therefore consume only from transfers. That is why a positive price of installed capital does not settle the insurance problem: capital may be valuable to its owners while workers have no direct income from it.

The AK form is deliberately austere. It does not say that all goods are produced by machines in a realistic economy. It gives a clean limiting case in which an arrival removes labour income without destroying the installed productive stock. That limit is useful for public finance because it separates two channels that otherwise move together: an increase in the value of capital and a decrease in the direct resources of workers. In a labour-using successor, higher productivity can raise both wages and returns. In the AK successor, only the second channel survives unless the government transfers some of the return.

The investment decision makes the ensuing transition explicit. A firm compares one extra unit of investment today with the induced growth in the capital stock and the shadow value q_t of installed capital. With a convex adjustment technology, the optimal investment rate is pinned down by the usual marginal condition

$$1 = q_t \Phi'(\iota_t). \tag{6.14}$$

Equations (6.13) and (6.14) describe a standard q-theory transition. A high successor price encourages investment, but it cannot change the capital stock at the date of arrival. The distinction is especially important for the tax. A continuation tax can lower the after-tax return and the price of installed capital immediately; its effect on the productive stock and on any future wage base occurs over time.

The firm distributes after-tax rents net of investment,

$$D_t = \big[(1 - \tau_t^K)A - \iota_t\big]K_t. \tag{6.15}$$

World pricing of the installed claim gives the familiar user-cost relation. With $\bar{r}^w$ denoting the world safe rate in efficiency units and q_t the price of a unit of installed capital,

$$(1 - \tau_t^K)A = \bar{r}^w q_t + \iota_t - \dot{q}_t - q_t[\Phi(\iota_t) - \delta]. \tag{6.16}$$

This equation does two jobs. It determines the after-tax rental return consistent with the selected price path, and it makes clear why a tax cannot be chosen independently of the price of installed equity. The government can formulate its continuation problem in terms of a tax path or in terms of an implementable allocation and price path; it cannot select tax rates, prices, and event gains as separate instruments.

For a stationary continuation, the price equation may have more than one positive algebraic root. The economically relevant solution is selected by the productive-value transversality condition: the present value of the installed claim must be generated by its future distributions rather than by an explosive terminal component. The appendix derives that selection. The main point is more basic. An AK successor needs an equilibrium price system and a tail condition even though its production function is simple.

For later use, it helps to write the government's account in levels before using the reduced representation. Let r_0^w be the pre-arrival world safe rate and let $\mathcal{R}_t^G$ denote the after-tax cash distribution on the public equity position, excluding the capital gain at the event. The source-tax term below is therefore not contained in $\mathcal{R}_t^G$. The public balance sheet satisfies

$$\dot{F}_t = r_0^w F_t + \mathcal{R}_t^G + \tau_t^K R_0^K(K_t)K_t - T_t. \tag{6.17}$$

The government may finance a transfer with current source-tax receipts, dividend income on the equity it owns, or a change in net financial wealth. These are distinct sources of funds. The first redistributes rental income. The second depends on an asset that had to be purchased at date zero or earlier. The third is limited by the debt and continuation restrictions. Combining them into F is convenient for solving the problem, but it should not obscure their different incidence and timing.

After arrival to successor j, the same accounting relation applies with successor objects:

$$\dot{F}_{j,t} = r_j^w F_{j,t} + \mathcal{R}_{j,t}^G + \tau_{j,t}^K R_j^K(K_{j,t})K_{j,t} - T_{j,t}. \tag{6.18}$$

In the AK successor, $W_{F,t} = 0$ and workers require $T_{F,t} > 0$ for strictly positive consumption. Equation (6.18) makes the substantive point concrete: a high successor value of capital helps workers only through the government's claim on the rents and the feasible continuation transfer policy.

The public budget is most transparent in reduced form. Before the arrival, suppressing the common growth normalization, it can be written as

$$\dot{e}_t = \bar{r}_0^w e_t + y_0(K_t) - \iota_t K_t - C_t^W - \psi_t \sum_j \lambda_j^* G_j, \tag{6.19}$$

where $y_0(K)$ is pre-arrival output net of the part already accounted for in the installed claim and

$$G_j \equiv \frac{q_j(K) - q}{q} \tag{6.20}$$

is the proportional total gain on installed equity in successor j. The reduced budget consolidates the public account with the equity-pricing equation. It does not say that source taxation has disappeared from the economy. The tax affects investment, prices, and the future distribution of income; it simply does not appear as a separate cash-flow term once those objects are consolidated.

At the arrival, K is continuous, while the reduced public balance sheet changes by the equity payoff:

$$e_j^+ = e + \psi G_j. \tag{6.21}$$

This is the event-time accounting identity used below. Safe debt has no direct successor-specific payoff. Nor does the predictable flow tax. Their effects operate through pre-arrival saving and through the continuation economy, respectively.

6.4 Equilibrium and the commitment Ramsey problem

Given a policy plan, a competitive equilibrium consists of household consumption and portfolio choices, firm investment and installed-equity prices, public transfers and balance-sheet positions, and foreign residual holdings such that: workers satisfy (6.3); owners satisfy their budget constraint and portfolio optimality conditions; firms satisfy (6.13) and (6.16); the government budget holds before and after the arrival; and equity and safe-asset positions clear with foreign investors taking the residual. The policy plan is adapted to information available at each date. In particular, the pre-arrival portfolio cannot depend on the successor that has not yet been realized.

The government solves a commitment Ramsey problem at date zero. It chooses an admissible contingent policy and the induced competitive equilibrium to maximize (6.4), taking inherited physical and financial states and protected claim values as given. The wording matters. The government is not handed a fresh balance sheet at the announcement date, and it does not confiscate previously valued positions by changing the continuation plan. But it may choose future taxes, transfers, and portfolio policies subject to the inherited claims, the public budget, world pricing, and the stated borrowing and solvency restrictions.

The AK case is useful because it separates two questions that are often run together. First, can the public acquire an event payoff that transfers resources to the labour-free economy? Equation (6.21) answers that question. Second, can the government use a source tax to finance workers once the economy has arrived in that regime? That depends on the continuation allocation, capitalization, protected claims, and the capital and debt constraints. A statement about the first question is not a theorem about the second.

More formally, a competitive equilibrium conditional on a policy plan has five blocks. The worker block gives consumption from wages and transfers. The owner block gives consumption and private portfolio positions subject to (6.6) and (6.7). The firm block gives investment, capital accumulation, and the price of installed capital. The public block gives taxes, transfers, debt, and equity positions subject to (6.17) and (6.18). Finally, the foreign block absorbs residual domestic equity and safe claims at world prices. The blocks are connected by the rental rate, installed-equity price, and the government budget. This list is useful because the event payoff is not a primitive object floating outside equilibrium: it is the difference between two prices that solve the firm and world-pricing blocks.

The commitment Ramsey problem closes the system by choosing the public block, not by directly selecting household consumption or successor equity prices. Given inherited capital, public wealth, gross claims, and protected owner values, the government chooses an adapted tax-transfer-debt-equity plan. A plan is admissible only if the competitive equilibrium just described exists and the policy respects the stated bounds in every successor. This is why the paper treats the public payoff map as a first step rather than as a decentralization theorem. The map says which event transfer can be purchased; the Ramsey problem says whether it belongs to a feasible optimal plan.

6.5 One successor and a complete event-payoff span

Suppose next that there is only one possible successor and that $G \neq 0$. Equation (6.21) becomes

$$e^+ = e + \psi G. \tag{6.22}$$

For any desired successor balance sheet e^+, the corresponding exposure is

$$\psi = \frac{e^+ - e}{G}. \tag{6.23}$$

Installed equity therefore spans the one-dimensional event innovation. This is a statement about the available public payoff direction. It does not mean that private households trade a complete set of securities, that every allocation is implementable, or that the allocation is first best. Feasibility still requires (6.11), positive successor consumption, protected claims, private solvency, and the selected continuation equilibrium.

Three different notions of completion are worth keeping separate. First, the statement here is complete *event-payoff span*: one nonzero payoff can move the one-dimensional uncertainty created by a single possible arrival. Second, complete private Arrow claims would mean that households can trade a full set of state-contingent claims for their own consumption. That is not assumed; workers remain outside private asset trade. Third, a first-best production-and-finance allocation would require the planner to choose production, redistribution, and insurance without the stated tax, ownership, debt, and protected-claim restrictions. The first property does not imply either of the other two.

The distinction has a practical implication. Even with a nonzero G, the exposure in (6.23) may violate $\psi \geq e$, may require a transfer path that cannot be financed after the arrival, or may conflict with a protected owner position. Conversely, incomplete private insurance does not imply that the public should take an arbitrarily large position. The public claim must still be valued at the world price and fitted into the government budget.

The one-successor case also clarifies the limited content of the source-tax result. On a promise-neutral, interior continuation with positive worker resources, world-price compatibility, and slack tax, debt, transfer, and portfolio constraints, the necessary conditions contain a cancellation that makes zero source tax a lower-bound candidate. This is not a version of the Chamley–Judd result. It relies on the one-dimensional event payoff, the small-open pricing closure, and the particular protection and continuation assumptions. If the lower bound is not optimal, if a constraint binds, or if a successor claim remains protected, the cancellation no longer determines the tax. The appendix gives the conditions and the KKT argument; no quantitative tax or welfare consequence follows here.

It is useful to see the state-price condition in components. Let V_e^+ be the marginal value to workers of one extra unit of successor public wealth and let μ_e be the pre-arrival shadow value of one extra unit of public wealth. The portfolio condition for a marginal increase in ψ is

$$\lambda V_e^+ G - \lambda^* \mu_e G = 0. \tag{6.24}$$

When $G \neq 0$, it reduces to

$$V_e^+ = \frac{\lambda^*}{\lambda}\mu_e. \tag{6.25}$$

The right-hand side is the world state price expressed in worker-value units. The equation does not say that the government likes a dollar in the successor exactly as a foreign investor does. It says that, when the only event risk is spanned and the portfolio choice is interior, the government cannot improve worker welfare by buying a little more or less of the available payoff at the prevailing price.

6.6 Two successor economies

Return now to the economically richer case. The arrival can lead either to partial automation, denoted P, or to full automation, denoted F. In the partial successor, labour remains productive and wages are positive under the stated technology. In the full successor, production is AK and workers depend on transfers. Let

$$\mathbf{G} = \begin{pmatrix} G_P \\ G_F \end{pmatrix}, \qquad \mathbf{e}^+ = \begin{pmatrix} e_P^+ \\ e_F^+ \end{pmatrix}. \tag{6.26}$$

Then the entire pre-arrival portfolio choice has the simple representation

$$\mathbf{e}^+ = e \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \psi \mathbf{G}. \tag{6.27}$$

Changing ψ moves successor wealth along one line in a two-dimensional space. It cannot generally select e_P^+ and e_F^+ independently.

One simple example fixes the geometry. Suppose equity loses ten percent in the partial successor and gains twenty percent in the labour-free successor, so that $\mathbf{G} = (-0.10, 0.20)'$.

Increasing ψ by one unit lowers public wealth in the partial successor by 0.10 and raises it in the full successor by 0.20. It cannot raise wealth in both successors, and it cannot preserve partial-successor wealth while raising full-successor wealth. The latter policy would require a second payoff vector that is not proportional to $\mathbf{G}$. The numerical values are only an illustration; the point is the dimension of the attainable set.

Proposition 6.1 (One installed-equity payoff)**.** *Fix the inherited state, successor technologies, continuation-policy map, and the predictable flow-tax closure. If* $(G_P, G_F) \neq (0, 0)$*, variation in the inherited installed-equity exposure spans only the one-dimensional set generated by* $\mathbf{G}$*. Safe debt and a predictable source tax add no direct successor-specific payoff at the arrival.*

The proposition is linear algebra, but it gives the central economic lesson of the event model. Domestic equity is not "insurance against automation" in the abstract. It is insurance for one particular package of resources across the two economies. If its large payoff comes when workers are well off in the partial successor and its small payoff comes when workers have no wages in the full successor, the public cannot turn it into a claim concentrated on the latter state merely by changing the size of the holding. An additional non-collinear payoff would be needed to choose two successor values separately.

The valuation condition describes how the government chooses along this restricted line. Let $V_{j,e}$ be the marginal value of successor public wealth for workers and let μ_e be the pre-arrival marginal value of public wealth. Define the successor-specific valuation residual

$$u_j \equiv \lambda_j V_{j,e} - \lambda_j^* \mu_e, \qquad j \in \{P, F\}. \tag{6.28}$$

At an interior portfolio choice on the stated continuation,

$$u_P G_P + u_F G_F = 0. \tag{6.29}$$

The government equates its marginal valuation of the available equity payoff to its world-price cost. Equation (6.29) does not imply $u_P = u_F = 0$. There may remain a component of worker fiscal need that is orthogonal to $\mathbf{G}$. That residual is the economic content of incomplete event-risk sharing.

Writing the same condition component by component makes the comparison with the one-successor case immediate:

$$\underbrace{(\lambda_P V_{P,e} - \lambda_P^* \mu_e)\, G_P}_{\text{partial successor}} + \underbrace{(\lambda_F V_{F,e} - \lambda_F^* \mu_e)\, G_F}_{\text{full successor}} = 0. \tag{6.30}$$

The public position balances the two weighted discrepancies. Either discrepancy may be nonzero. For example, the labour-free successor can have a high worker value of public resources because workers have no wage income, while the equity payoff may be small or negative there. The portfolio can then be at its optimum along the available line even though the government would value an additional claim paying only in that successor. Equation (6.30) is a condition for the chosen position, not a claim that every successor-specific insurance need has been priced away.

There are three reasons not to read the projection condition as a policy prescription. First, it is an interior condition; a borrowing limit, a protected claim, a transfer floor, or a position restriction adds a multiplier. Second, successor values include the response of the continuation economy, including the capitalization effect of future source taxes. Third, the condition prices a given payoff menu. It does not prove that the menu is rich enough to finance worker consumption in either successor.

6.7 What the event model adds

The two-successor economy complements the continuous-risk model. In both cases the relevant object is the alignment between a traded payoff and worker-relevant public resources. The discrete model makes a further point visible: transformative uncertainty is often uncertainty over successor production regimes, not merely over the next realization of a common shock. An asset can be valuable in both successor economies and still leave the public unable to move resources in the direction that matters most for workers.

The analysis is deliberately narrower than a quantitative policy exercise. It derives the event-time payoff map, the rank of the available equity payoff, the competitive equilibrium closure, and a conditional state-price projection. It does not establish a global transition, an optimal source-tax path, or a welfare ranking between the two successor economies. Those conclusions require a solved continuation and a feasible public balance sheet for every relevant history.

7 A fixed-policy transition account

The Ramsey economies answer a normative question: which state-contingent taxes, transfers, and public positions does the government choose when it can commit at date zero? A different question is often useful before solving that problem. What does a specified fiscal rule do when factor incomes follow a stated automation transition? This section records a deliberately reduced-form exercise that answers the latter question. It is an accounting and incidence device, not a second optimization model.

The distinction is more than terminology. The exercise fixes aggregate capital, domestic private ownership, tax rates, and the government rule for transfers, saving, and domestic equity. It prices domestic equity with an exogenous world stochastic discount factor and lets foreign investors absorb the residual supply. Holding those objects fixed is useful because it shows how a rule transmits a change in factor incomes through the public balance sheet. It also means that the exercise cannot identify the value of Ramsey reoptimization, an optimal public portfolio, or the capital-deepening mechanism in Moll et al. (2022).

7.1 A portable wage identity

Let a index the automation state. The fixed-policy exercise uses a reduced-form factor-income schedule

$$\log\left(\frac{Y}{\bar{Y}}\right) = z + g(a) - g(\bar{a}), \qquad W = (1-a)Y, \tag{7.1}$$

where z is an aggregate productivity disturbance and $g(a)$ captures the scale effect associated with automation. Differentiating gives

$$\frac{\mathrm{d}\log W}{\mathrm{d}a} = g'(a) - \frac{1}{1-a}. \tag{7.2}$$

The identity separates two forces. Automation raises output through the reduced-form scale term $g'(a)$. At the same time, it lowers labour's share mechanically. Wages fall with a only when the scale effect is weaker than the displacement term $1/(1-a)$. This is a useful reminder for the evidence in Section 1.1: a declining labour share is not by itself evidence that workers' real resources have fallen, and a rise in output is not by itself evidence that they have risen.

Equation (7.2) is not a structural production result. Fixed capital means that $g(a)$ absorbs the capital-deepening and task-reallocation mechanisms that a richer model would determine endogenously. The schedule is chosen to organize selected factor-income movements, not to replicate the technology of any one historical episode.

Capital income is the complementary object in this accounting exercise. With the capital share equal to a, it is $R = aY$. An increase in a therefore changes both the division of a given output level and the output level itself through $g(a)$. The model does not infer this relationship from micro data. It takes the path for (a, z) as an input and asks how an announced fiscal rule distributes its consequences. That limited purpose is valuable: it prevents the public-accounting exercise from being mistaken for an empirical model of technical change.

7.2 What the announced rule holds fixed

The government announces a rule $p = (F^\star, \kappa_S, \omega_D)$. The rule specifies a reference public fund $F^\star$, the speed κ_S at which saving responds to the gap from that reference, and the share ω_D of the fund held as domestic equity. For a given realization of (z, a), the real side maps into wages and capital income; the rule maps these incomes and world-priced returns into transfers and public wealth; and the resulting group-specific consumption paths map into welfare. The government does not solve a new first-order condition after a shock.

One can write the rule in a simple flow form. Let S_t^p be public saving under rule p and let F_t be public net wealth. A representative version is

$$S_t^p = \kappa_S(F^\star - F_t), \qquad \Theta_t^D = \omega_D F_t, \tag{7.3}$$

where Θ_t^D is the market value of domestic equity held by the government. A transfer rule closes the public budget by allocating current tax and asset income net of saving between workers and owners. The exact transfer shares are part of the announced rule. They are not chosen after a realization of automation, and they are not implied by the ownership share ω_D.

This representation makes two mechanical properties visible. When $F_t = 0$, the portfolio share has no contemporaneous effect because the government has no fund to allocate between assets. And when the government raises its saving target while holding tax rates fixed, the initial transfer envelope must shrink. Any later gain from a larger fund has to compensate for that initial use of resources. These are accounting facts, not claims about the desirability of a particular target or asset mix.

This formulation gives public domestic ownership a transparent accounting treatment. Let D^G be dividends received by the government and let R_{tax} be tax revenue. The public cash return from its domestic position is

$$R^G_{\text{cash}} = R_{\text{tax}} + (1 - \tau_K)D^G. \tag{7.4}$$

The tax and the after-tax dividend are distinct components. Counting the full dividend as both tax revenue and an asset payoff would create resources that no agent receives. Likewise, a comparison of public portfolios begins with a fair date-zero swap: public wealth is held fixed and the government exchanges safe assets for domestic equity at the world price. Foreign investors take the other side. No comparison receives a free initial revaluation.

The same accounting clarifies saving. On a slack reference path, one additional unit of desired public accumulation must reduce the direct transfer envelope by one unit before return effects. A larger public fund is therefore not an endowment to workers. It is a different intertemporal allocation of their resources. This is precisely why the fixed-rule exercise is a useful complement to the Ramsey model: it makes visible the financing margin that can disappear when a public balance sheet is described only by its terminal value.

There is a useful consolidated-account check. Add the public account to domestic private income and cancel taxes and transfers that occur between domestic agents. What remains is domestic output net of investment, the safe return on net foreign assets, and the after-tax dividend paid to foreign residual investors, together with the revaluation of claims. The check prevents a public purchase of domestic equity from being treated as a new domestic resource. It changes who receives future dividends and who bears their risk. It does not create the dividend.

7.3 Evaluating a rule is not choosing a rule

The welfare equation is a linear evaluation equation under the announced rule. If c_i^p is consumption for group $i \in \{W, K\}$ and $\mathcal{L}^p$ is the generator induced by that rule, then the continuation value satisfies

$$\rho_i V_i^p = u_i(c_i^p) + \mathcal{L}^p V_i^p. \tag{7.5}$$

When the automation path is explicitly time varying, the corresponding equation also includes $\partial_t V_i^p$. There is no maximization over p in (7.5). Changing the rule changes the generator, the transfer path, and the price process that must be used to value the new path.

This distinction rules out a tempting comparison. One cannot interpret the difference between this exercise and the Brownian Ramsey economy as the value of policy reoptimization. The two environments differ in production, household portfolios, ownership, fiscal instruments, and pricing closure. The appropriate reoptimization comparison would keep the same Brownian economy and the same inherited state fixed, then compare continuation under an inherited rule with continuation under the Ramsey plan. That comparison is not supplied by the reduced-form exercise.

Nor should worker and owner values be silently aggregated. A transition can raise both values, lower both, or change their ranking under a stated transfer rule. Those are incidence statements. A social welfare comparison requires explicit interpersonal weights and an accounting for foreign residual investors.

The appropriate counterfactual also holds the comparison itself fixed. A transition effect compares the same announced rule with and without the specified automation path. The public fund, tax rates, transfer-feedback coefficients, initial wealth, ownership, and world-pricing convention must be identical at the initial date. Re-anchoring the transfer intercept after changing the transition would silently introduce a new rule. Similarly, comparing two portfolio shares requires the same initial fund and a fair exchange at the initial equity price. These conventions do not make the exercise more realistic; they make its reported incidence interpretable.

7.4 Expected reversion and balance-sheet risk

The exercise also provides a warning about rule design. A feedback rule can make the expected public fund mean reverting while leaving its risk uncontrolled. In a scalar approximation, let the drift of the fund be

$$a_F = r^w + \omega_D \beta_D - (\phi_L + \phi_K), \tag{7.6}$$

where r^w is the world safe return, β_D is the expected excess return on domestic equity, and $\phi_L + \phi_K$ summarizes transfer feedback. Mean reversion requires $a_F < 0$. If the risky position is proportional to the fund, however, a finite second moment additionally requires

$$2a_F + \omega_D^2 \|\sigma_D\|^2 < 0. \tag{7.7}$$

The second condition is stronger. A rule can push expected wealth back toward target and nevertheless amplify the dispersion of fiscal outcomes because the risky exposure grows with the fund itself.

The distinction can be seen from a stylized fund law,

$$\mathrm{d}F_t = a_F F_t \ \mathrm{d}t + \omega_D F_t \sigma_D \ \mathrm{d}Z_t. \tag{7.8}$$

Taking expectations gives $\mathrm{d}\mathbb{E}F_t/\,\mathrm{d}t = a_F\mathbb{E}F_t$, so $a_F < 0$ is enough for the mean to return. Applying Itô's formula to F_t^2 gives a drift $(2a_F + \omega_D^2\|\sigma_D\|^2)F_t^2$, which yields (7.7). The extra positive term is the variance generated by a position that scales with the fund. A policymaker who inspects only expected wealth can therefore miss a growing probability of very weak public resources.

This is not an optimal-portfolio condition. It is a stability check for a particular proportional rule. Binding transfer, wealth, or ownership limits change the dynamics, and the full multi-dimensional condition uses the corresponding generalized second-moment operator. The implication is nevertheless straightforward: a target for expected public wealth is not a definition of fiscal capacity. A rule intended to support workers in bad states must also control the lower tail of the balance sheet.

7.5 What the current calculations say

The available calculations are intentionally modest. Under the zero-price-of-risk version of the announced rule, the reported effects of changing the domestic-equity share are below the predeclared 10^{-4} consumption-equivalent reporting threshold. They are therefore numerically uninformative about portfolio composition. Coarse transition calculations report positive welfare changes for workers and owners under the imposed transition and transfer rule, with larger gains for owners. Those signs describe the model's stated rule; they do not identify a desirable fund, portfolio, or tax system.

An attempted nonzero-price-of-risk sensitivity changes continuation pricing without replaying the within-horizon public account under the altered price process. Its welfare values cannot be used. For the same reason, this section reports no portfolio figure and no welfare surface. The useful conclusion at present is negative but substantive: an apparently sensible domestic-equity rule need not generate a detectable effect when its payoff is weakly aligned with the public exposure it is meant to hedge. The numerical appendix records the accounting checks, the coarse transition exercise, and the conditions required before a quantitative comparison can carry more weight.

This reporting choice is substantive. A number that is smaller than the reporting threshold cannot tell the reader whether the rule has a small beneficial effect, a small harmful effect, or no economically material effect. The nonzero price-of-risk exercise has a different problem: its objects are not comparable because the changed price process was not carried through the public cash account. Neither limitation is repaired by a smoother figure. The appropriate use of the current calculations is to show the accounting architecture and to record a recognized null, not to rank public asset portfolios.

The discipline is the same for a future positive result. A larger effect would not be more persuasive merely because it is larger. It would need the same initial public wealth, a complete replay of prices and fiscal cash flows, and numerical uncertainty that is small relative to the reported consumption-equivalent difference.

7.6 What the fixed rule isolates

With policy held fixed, the exercise isolates the accounting link between factor shares, ownership, and the fiscal distribution. Equation (7.2) separates scale from displacement; the consolidated budget keeps public ownership from becoming a fictitious resource; and the moment conditions distinguish a stable mean path from a stable fiscal distribution. These are incidence and accounting results under an announced rule, not a second Ramsey answer.

The broader lesson is the same as in the previous sections. A public portfolio has to be evaluated by the payoffs it delivers, the resources it displaces, and the constraints under which it is financed. Holding a policy fixed is often the cleanest way to see those accounting links. It is not a substitute for choosing the policy.

8 Discussion and conclusion

The paper begins from a simple observation. Automation can raise output and the value of capital at the same time that it weakens the resources on which workers and the Treasury rely. The public-finance question is not whether capital has a high expected return. It is whether the assets, taxes, and borrowing arrangements available to the government pay when worker consumption is scarce. That question links the distributional concerns emphasized by Moll et al. (2022) to the public portfolio problem studied by Bhandari et al. (2017) and Aparisi de Lannoy et al. (2025).

The answer is necessarily conditional on the economic environment. In the Brownian model, the government compares the worker value of a fiscal payoff with its financial price and chooses the available payoff projection. In the event-arrival model, it chooses a position in installed domestic equity before knowing which successor economy will arrive. In both cases, an asset's label is not enough. The sign and size of its value for workers depend on the alignment of its payoff with wages, tax resources, transfers, and the continuation constraints of the public balance sheet.

8.1 The insurance problem is about payoffs, not asset classes

It is tempting to reason from a simple portfolio slogan: if automation raises the value of capital, the government should own capital. The paper shows why that slogan is incomplete. A public position has two conceptually distinct components. One is ordinary demand for a compensated return. The other is a hedge against the component of worker-relevant fiscal resources that covaries with the asset payoff. These components can be large and offsetting. A small net holding may hide substantial return demand and hedging demand; a large holding need not be much insurance.

The distinction becomes sharper when there are more risks than traded public claims.

With one risky payoff and two independent Brownian innovations, the government can offset only the projection of fiscal exposure onto that payoff. The orthogonal component remains uninsured. With one installed-equity payoff and two successor economies, the government can move public wealth only along one line in the pair of successor values. These are not merely technical rank conditions. They explain why an attractive asset can fail to insure the state in which workers need support most.

This perspective also disciplines claims about a public wealth fund. Public financial net worth, the value of resources relevant to workers, and pledgeable fiscal capacity are different objects. A future wage bill can matter greatly for worker welfare and yet be unavailable as collateral. A high expected return can raise the present value of an asset without increasing the safe amount the government can promise in every state. And a portfolio that pays in a bad worker state may still be infeasible because the position violates a debt limit, an inherited claim, a transfer floor, or private solvency.

8.2 Capital taxation is an intertemporal instrument

The capital-tax results fit naturally into this payoff perspective. A source tax redistributes current rental income, but it also changes the after-tax return to installed capital and therefore the future capital stock, wages, and tax base. It is not an event-time security. A predictable rental-flow tax cannot manufacture a payoff at the instant an event occurs, even if successor tax rates can later alter prices and investment.

This timing explains why neither a zero-tax theorem nor a confiscation argument is a good shortcut for the present setting. The conventional long-run capital-tax logic turns on a different set of assumptions about preferences, production, fiscal instruments, and the relevant transition. The log-utility calculation here can produce a long-run tax formula under its stated interior assumptions, but it does not select a universal rate. In the common-discount special case, the formula yields one half; that number is a property of the model's patient workers and owners, not a policy target. The older result associated with Farhi (2010) is informative about incomplete markets and ownership, but it does not collapse the public payoff problem into a general zero-tax conclusion.

The contrast with the AK successor makes the point sharper. In an AK economy, the return to installed capital is central to output and there is no wage bill to protect directly. A source tax can transfer part of the rental flow to workers, but it also changes the value of the installed claim and the incentive to accumulate capital. In a labour-using economy, the same tax has an additional effect through the future wage bill. These are different continuation problems even when both are described as "capital taxation." The relevant comparison holds fixed inherited capital, protected claims, and the world-pricing closure; otherwise a change in a tax rule can be confused with a change in the public wealth inherited at the arrival.

Public ownership is not by itself sufficient to resolve this trade-off. Owning domestic equity gives the government a claim on dividends and revaluations. It does not tell the government how much of that claim can be converted into transfers without violating debt, tax, or continuation constraints. Nor does it create a separate event payoff for each successor economy. The public owner faces the same q-theory and price-capitalization relations as every other owner, while the government additionally faces a distributional objective. That is why public ownership belongs in a joint tax, debt, and portfolio problem rather than in a stand-alone nationalization argument.

The transition is equally important. At impact, installed capital is inherited and the tax rate may be constrained to move gradually. A persistent automation disturbance can initially reduce wages, then raise them later if capital accumulation responds. Taxing a temporary quasi-rent can provide resources while the inherited stock remains in place; it can also depress the accumulation that eventually restores worker income. The forward-looking tax condition therefore compares current redistribution with the worker value of future capital. It is not a static Laffer calculation, and its sign cannot be read from the current rental-tax base alone.

This is one reason to separate source taxation from safe finance and risky public claims. Safe debt reallocates resources across dates but has no state-specific payoff. Risky claims reallocate resources across the states in which they pay. Taxes alter the distribution and accumulation process that generates future income. Their interaction is central, but they are not substitutes.

8.3 Successor uncertainty and weak links

The event model is meant to be read alongside recent macroeconomic work on advanced AI, not as a forecast that a labour-free economy is imminent. Acemoglu (2025) emphasizes that the aggregate effects of current AI depend on task exposure and the scope for productivity gains. Jones and Tonetti (2026) emphasize weak links that can limit a growth explosion even when individual technologies improve rapidly. Work on transformative scenarios similarly distinguishes alternative paths for capabilities, adoption, and economic organization rather than assuming a single deterministic transition (Korinek and Suh, 2024; Trammell and Korinek, 2026).

The weak-link perspective is especially relevant for tax policy. If a scarce input, energy constraint, complementary task, or institutional bottleneck limits the expansion of capital services, a high measured return on a subset of AI-exposed assets need not translate into a permanently high broad tax base. Conversely, if a bottleneck is relaxed and capital accumulation becomes rapid, a tax that extracts more current rent can have a larger effect on the future stock that supports output and wages. The model does not choose between these technological possibilities. It shows why a fiscal plan should be conditioned on them rather than extrapolated mechanically from current valuations.

The two-successor construction translates that observation into a public-finance problem. The partial and full successors differ in the existence of wage income, the return to installed capital, and the fiscal resources required to support workers. A portfolio chosen before the arrival is exposed to both. The key limitation is not that the government lacks a view about the probability of each successor. It is that one traded equity payoff does not generally let the government select a separate fiscal position in each one.

This distinction also helps interpret the growing evidence on asset prices and AI. Market valuations can reveal investors' assessment of future profits, risk, and discount rates. They need not reveal how a public balance sheet should insure labour income. For example, Andrews and Farboodi (2025) study whether financial markets appear to price transformative AI. The object in this paper is different: the wedge between the valuation of a payoff by the world investor and its value for worker consumption and public transfers. Evidence on valuations is therefore useful input, but not a solution to the insurance problem.

8.4 Institutions, governance, and capacity

The paper's policy implications are institutional before they are quantitative. A government considering an automation-linked public portfolio needs a precise mapping from asset payoffs to the worker and fiscal states it wants to insure. It needs an explicit financing rule for the initial position and for losses. It needs to specify who owns domestic capital, who receives transfers, and which inherited claims are protected. Finally, it needs a borrowing and solvency rule that remains meaningful after the states in which the hedge pays poorly.

Those requirements are not bureaucratic add-ons. They determine the economics. A fund can look large because it has borrowed against a volatile asset; its gross equity position then says little about safe fiscal capacity. A tax can look powerful because it is applied to a large capital stock; its continuation effect may nevertheless lower the future wages and rents that matter for workers. A promise to distribute capital returns can insure workers only if the public actually owns the payoff and can sustain the promised transfer rule when returns disappoint.

This emphasis is close in spirit to the public-portfolio literature, which treats debt management and asset policy as part of the same contingent fiscal plan rather than as separate accounts (Bhandari et al., 2017; Aparisi de Lannoy et al., 2025). The automation setting adds a distributional reason to make the payoff map explicit. Workers may be shut out of the private portfolio that benefits from automation, while owners and foreign investors hold claims on the resulting rents. The relevant public portfolio is therefore not a representative-agent investment problem.

Governance matters for the same reason. A credible public fund needs a mandate that states which risks it is meant to insure, who is eligible for its transfers, what debt can support its positions, and how its rules change after an arrival. Without those objects, it is impossible to distinguish insurance from a leveraged directional bet or from a transfer to incumbent asset holders. The model does not prescribe an institution, but it indicates what an institution would have to disclose for its policy to be evaluated: exposures, funding sources, protected commitments, and the successor states in which its transfers remain feasible.

8.5 What the historical evidence can and cannot do

The United States and German episodes in the opening section are deliberately descriptive. They show how a falling labour share, rising equity values, and a changing tax composition can coexist, and why the associated fiscal question cannot be settled by an aggregate output series. The U.S. evidence over 1990–2007 and the German evidence over 1994–2014 are not causal estimates of automation. The shorter German episode around 2004–07 is especially useful as an accounting illustration: a worker squeeze and a capital boom need not coincide with an immediate Treasury crisis.

That scope is appropriate. The model asks for state-contingent exposures, while the historical figures provide only broad co-movements. A quantitative policy design would need evidence on worker earnings risk, the tax base, the price and payoff of candidate claims, foreign ownership, and the constraints governing public borrowing and transfer policy. It would also need a credible way to distinguish automation from demand, trade, monetary conditions, and other changes that move wages and asset prices together. The paper does not supply that empirical identification.

The United States and Germany also illustrate why the paper does not treat a capital boom as a fiscal windfall. In the United States between 1990 and 2007, a fall in the labour share coexisted with rising aggregate and hourly compensation and strong equity returns. The observation is consistent with scale effects and capital gains offsetting some displacement; it does not identify their causes. In Germany between 1994 and 2014, the labour share declined while per-worker compensation was much weaker. The 2004–07 episode is a particularly useful warning: capital-tax-base proxies and domestic equity prices rose, corporate-tax receipts improved, and yet the resulting evidence does not show that a government had acquired a permanent insurance resource. Transfers, interest costs, business-cycle conditions, and the composition of ownership all matter.

The descriptive episodes should therefore be read as a map of coexistences, not a calibration target. They motivate the distinction between wage resources, fiscal flows, and priced capital claims. They cannot determine the covariance structure needed for a public portfolio or establish that observed changes were caused by automation. A useful empirical programme would combine household earnings data, tax administrative data, firm ownership, and asset prices at a common frequency, then ask whether the candidate public claim pays in the same states as worker income falls.

8.6 Limits and next steps

Several limits are material. The Brownian model is a mechanism laboratory, not an empirical estimate of an automation process. The small-open event model takes world pricing as given and abstracts from feedback from domestic policy to global asset prices. The labour-free AK successor is a polar case chosen to make the worker-income problem transparent; it is not a prediction. The fixed-policy calculations evaluate an imposed rule and currently provide no resolved portfolio comparison.

There are also theoretical limits. The payoff-rank and valuation-projection results do not prove global existence, select every possible continuation, or determine fiscal capacity. The source-tax characterizations are conditional on the stated protection, pricing, and constraint assumptions. In particular, a complete event-payoff span with one successor does not imply complete private insurance or a first-best allocation. These qualifications matter because relaxing them can change the instrument set and the sign of a policy response.

The natural next empirical step is not to infer an optimal fund from an asset-return regression. It is to estimate the joint payoff map: how worker income, tax revenue, candidate public claims, and borrowing capacity move across the shocks and successor scenarios that are economically relevant. The natural theoretical step is to solve richer continuation economies under explicit institutional restrictions and to add independent claims only where their payoffs genuinely enlarge the public span.

8.7 Conclusion

Automation raises a public-insurance problem because it can change the distribution of income and the value of private capital at the same time. The government cannot answer that problem by asking only whether it should tax capital more or own more equity. It must ask which payoffs it can buy, when those payoffs arrive, which worker and fiscal states they insure, and whether the resulting balance sheet is feasible.

The models in this paper make that discipline explicit. In continuous time, the public portfolio can hedge the component of fiscal exposure that lies in its payoff span when the corresponding position is feasible. Before an arrival with several possible successor economies, one installed-equity position buys one vector of successor resources, not a separate insurance payment for each economy. Capital taxation, safe finance, and public asset holdings then operate on different margins of the same fiscal plan. The payoff map, rather than the name of an asset or the height of its expected return, is the right starting point for designing public insurance against automation risk.

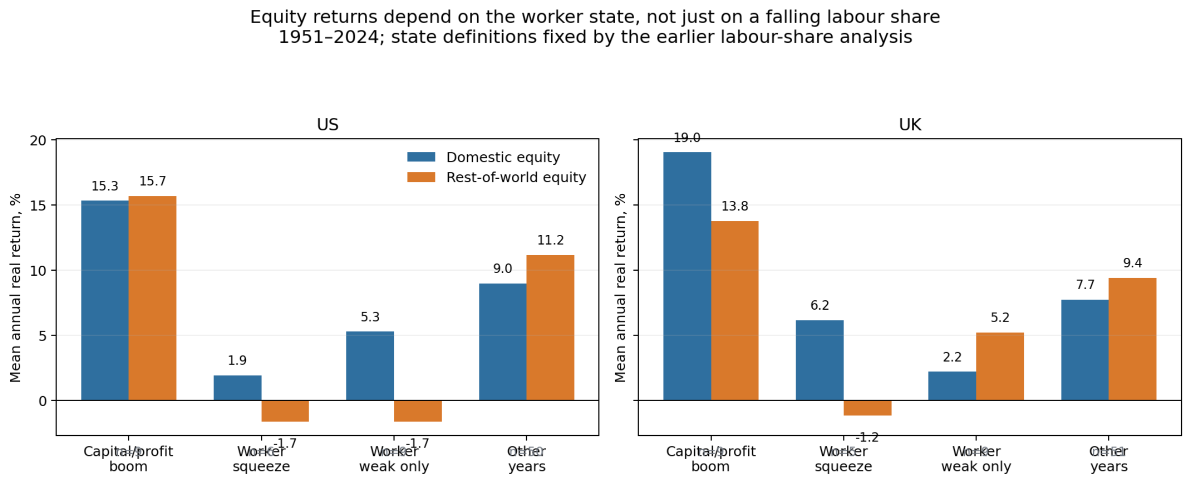

Figure 5: Asset returns across provisional worker and fiscal states. The figure summarizes descriptive state cells rather than a simulated model output. The small cells make the figure descriptive: it motivates measuring the joint worker, fiscal, payoff, and funding state rather than treating domestic equity as a reliable hedge.

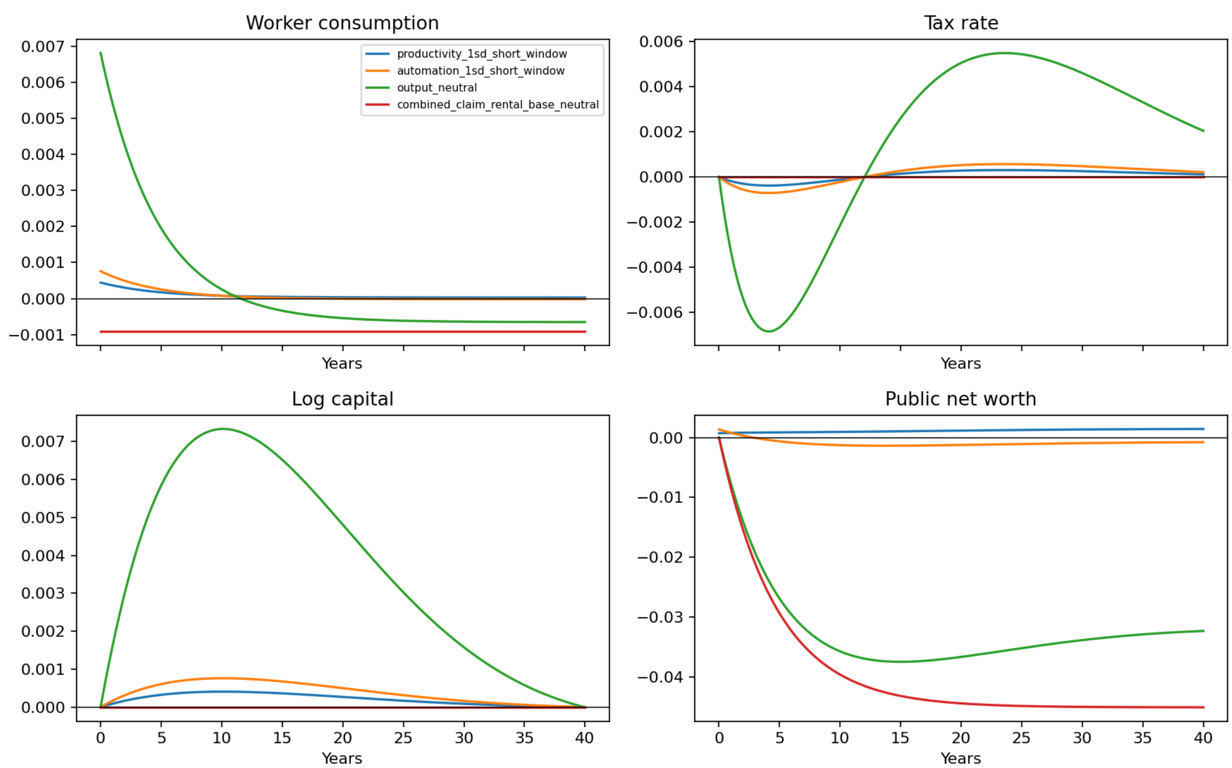

Figure 6: Selected local transition paths. The figure reproduces development-stage code output from the local quadratic approximation around the illustrative deterministic anchor. Responses are deviations in the units shown by the source output. The paths are mechanism illustrations, not calibrated estimates or accepted quantitative policy effects; matched nonlinear and boundary-slack validation remains outstanding.

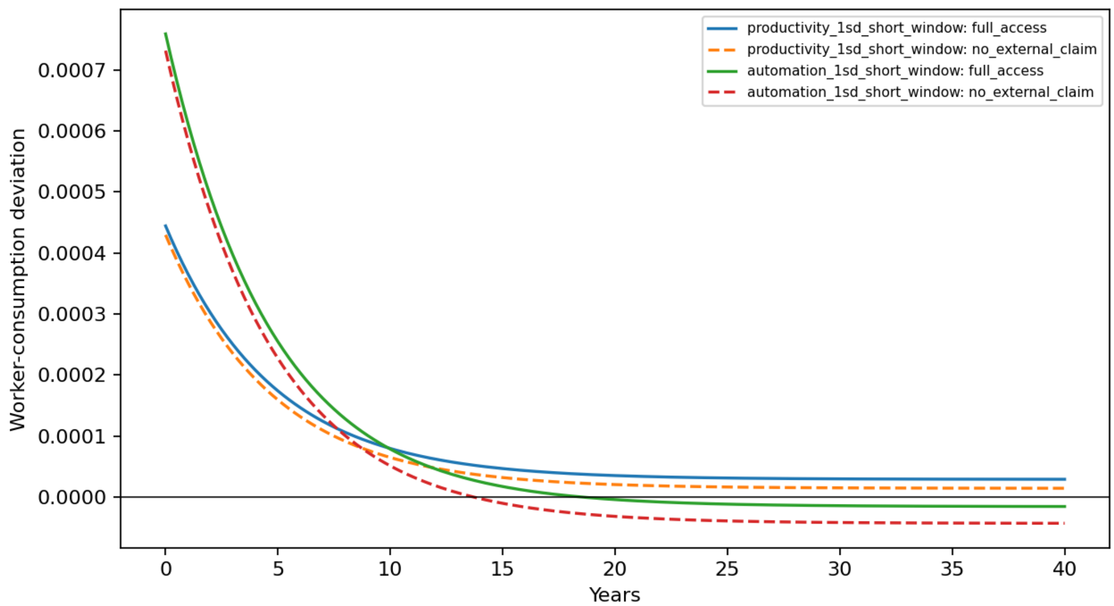

Figure 7: Worker-consumption responses with and without the external claim. The figure is direct development-stage output from the same illustrative local solution as Figure 6. It is not evidence of an empirically meaningful welfare gain. Its role is to display the common-state counterfactual and the timing of the consumption difference generated by claim access.

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